

ESG Momentum and the Business Cycle: Evidence from European Equity Markets

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Abstract

This thesis examines whether ESG momentum, which is measured as the 12-month Re-finitiv ESG score change of a firm, is positively associated with stock returns among stocks in the STOXX Europe 600 index between June 2016 and September 2025. To do so, it uses pooled OLS, random effects, fixed effects, and two-step difference generalized method of moments estimation. It further investigates whether long–short ESG momentum portfolio payoffs are related to the business cycle and whether the common macroeconomic variables dividend yield, short-term yield, term spread, and default spread are associated with ESG momentum payoffs. The results suggest that ESG score levels are generally negatively related to stock returns and ESG momentum is positively associated with returns, although the significance of the evidence is subject to model specifications. This result is consistent with the interpretation that ESG levels and ESG momentum capture different economic mechanisms. High ESG score levels may proxy for lower risk, while ESG momentum may capture repricing during the period in which new information about ESG activities is incorporated into stock prices. The business cycle analysis further suggests that ESG momentum payoffs are unlikely to be purely firm-specific. ESG momentum portfolios with high ESG score changes exhibit higher returns predicted by macro variables, and payoffs become small or negative after removing the macro-predicted component, especially in value-weighted ESG momentum portfolios. Lagged macroeconomic variables appear to account for a meaningful share of payoffs, as direct payoff regressions suggest that ESG momentum returns are partly explained by dividend yield, short-term yield, term spread, and default spread. However, the evidence remains sensitive to model specifications and weighting schemes.

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List of Abbreviations

AAA Highest credit rating category

AR Autoregressive

BAA Moody's Baa corporate bond rating category

CAPM Capital Asset Pricing Model

CDP Carbon Disclosure Project is a global non-profit organization that runs an environmental disclosure system.

CEPR Centre for Economic Policy Research

CMA (Conservative Minus Aggressive) is the average return on the two conservative investment portfolios minus the average return on the two aggressive investment portfolios.

COVID-19 Coronavirus Disease 2019

CSR Corporate Social Responsibility

DEF Default Spread defined as the difference between the average yield of corporate bonds rated BAA by Moody's and the average yield of corporate bonds with a Moody's rating of AAA.

DIV Dividend yield on the market defined as the total dividend payments accruing to the STOXX Europe 600 index over the previous 12 months divided by the current level of the index.

DMI Domini Social Index 400 (Now known as MSCI KLD 400 Social Index)

EABCN Euro Area Business Cycle Network

ECB European Central Bank

ES Environmental and Social

ESG Environmental, Social and Governance

FE Fixed Effects Model

FF-3 (Fama-French Three Factor Model) The Fama/French factors are constructed using the 6 value-weight portfolios formed on size and book-to-market.

FF-5 (Fama-French Five Factor Model) The Fama/French 5 factors (2x3) are constructed using the 6 value-weight portfolios formed on size and book-to-market, the 6 value-weight portfolios formed on size and operating profitability, and the 6 value-weight portfolios formed on size and investment.

FTSE Financial Times Stock Exchange

GICS Global Industry Classification Standard

GLS Generalized Least Squares

GMM Two-Step Difference Generalized Method of Moments

HAC Heteroskedasticity and Autocorrelation Consistent

ICB Industry Classification Benchmark

ISS Institutional Shareholder Services is an advisory firm providing data and analytics.

KLD (Kinder, Lydenberg, Domini & Co.) refers to KLD Research & Analytics, Inc., a firm in ESG ratings founded by Amy Domini. It was acquired by RiskMetrics in 2009. MSCI bought RiskMetrics in 2010.

LSEG London Stock Exchange Group

MSCI Morgan Stanley Capital International

MSCI ESG KLD STATS is an annual dataset that applies ESG performance indicators to a universe of publicly traded firms.

NBER National Bureau of Economic Research

NYSE New York Stock Exchange

OLS Ordinary Least Squares

P1 Portfolio 1 (lowest ESG momentum portfolio)

P5 Portfolio 5 (highest ESG momentum portfolio)

POLS Pooled Ordinary Least Squares Regression

RE Random Effects Model

S&P 500 (Standard & Poor's 500 Composite Stock Price Index) is a market-capitalization-weighted, float-adjusted stock market index that tracks the performance of 500 large publicly traded companies in the United States of America.

TERM Term Spread defined as the difference between the annualized 10-year euro area AAA government bond spot rate and the annualized 3-month euro area AAA government bond spot rate.

U.K. United Kingdom

U.S. United States

VIF Variance Inflation Factor

YLD Yield defined as the annualized 3-month euro area AAA government bond spot rate.

List of Symbols

a_i	Time-invariant individual effect of stock i
$\bar{a}_i$	Time average time-invariant individual effect of stock i
c_t	Scale factor in the forward orthogonal deviations transformation
d	Day
g	Integer controlling how much serial correlation is allowed in computing the standard errors
i	Stock
k	Degrees of freedom of the between regression
m	Month
n	Number of monthly observations
q	Quarter
$r_{i,m}^{ADJ}$	By macroeconomic variables unexplained part of returns
$\hat{r}_{i,m}$	Predicted return of stock i in month m
$r_{i,m}$	Return of stock i in month m
s	Future time index in the forward orthogonal deviations transformation
t	Period
$\bar{u}_i$	Time average idiosyncratic error of stock i
$u_{i,q}$	Error term of stock i in quarter q
$u_{i,t}$	Idiosyncratic error of stock i in period t
u_{m-1}	Error term in the rolling payoff regression
$v_{i,t}$	Composite error of stock i in period t
$w_{i,d}$	Relative free-float market capitalization of stock i on day d , capped at 20%
x_i	Full regressor history of stock i
$\bar{x}_i$	Time average of the regressors for stock i
$x_{i,t}$	$K \times 1$ vector of regressors

$\bar{y}_i$	Time average of the dependent variable for stock i
$y_{i,t}$	Return of stock i in period t
$y_{i,t-1}$	By one period lagged return of stock i
β	Coefficient vector
β_0	Intercept
β_1	Coefficient on the industry dummy variable
β_{10}	Coefficient on the 12-month ESG score change
β_{11}	Coefficient on the time dummy variables
β_2	Coefficient on the country dummy variable
β_3	Coefficient on the CAPM beta coefficient
β_4	Coefficient on size
β_5	Coefficient on value
β_6	Coefficient on profitability
β_7	Coefficient on investment
β_8	Coefficient on price momentum
β_9	Coefficient on the ESG score
$\beta_{DEF,m}$	Coefficient on default spread in the rolling payoff regression
$\beta_{DIV,m}$	Coefficient on dividend yield in the rolling payoff regression
$\hat{\beta}_{DEF,m}$	Estimated coefficient on default spread used to predict ESG momentum payoffs
$\hat{\beta}_{DIV,m}$	Estimated coefficient on dividend yield used to predict ESG momentum payoffs
$\hat{\beta}_{TERM,m}$	Estimated coefficient on term spread used to predict ESG momentum payoffs
$\hat{\beta}_{YLD,m}$	Estimated coefficient on yield used to predict ESG momentum payoffs
$\beta_{TERM,m}$	Coefficient on term spread in the rolling payoff regression
$\beta_{YLD,m}$	Coefficient on yield in the rolling payoff regression
$Beta_{i,q}$	CAPM beta coefficient of stock i in quarter q
$Country_i$	Country dummy variable of stock i

DEF_m	Default spread in month m
DEF_{m-2}	By two months lagged default spread
DEF_{m-1}	By one month lagged default spread
$D_{i,d}$	Dividend payment of stock i on day d
DIV_m	Dividend yield on the market in month m
DIV_{m-2}	By two months lagged dividend yield on the market
DIV_{m-1}	By one month lagged dividend yield on the market
DP_m^{TTM}	Sum of dividend points over the previous 12 months
$\Delta ESG_{i,q}$	12-month ESG score change of stock i in quarter q
$\Delta ESG_{i,t}$	ESG momentum of stock i in period t
$ESG_{i,q}$	ESG score of stock i in quarter q
$ESG\ MOM_m^{ADJ,ew/vw}$	Equal-weighted or value-weighted ESG momentum adjusted payoff in month m
$ESG\ MOM_m^{ADJ}$	ESG momentum adjusted payoff in month m
$ESG\ MOM_m^{ew}$	Equal-weighted ESG momentum payoff in month m
$ESG\ MOM_m^{ew/vw}$	Equal-weighted or value-weighted ESG momentum payoff in month m
$ESG\ MOM_{m-1}^{ew/vw}$	Equal-weighted or value-weighted ESG momentum payoff in month $m - 1$
$\widehat{ESG\ MOM}_m$	By macroeconomic variables predicted ESG momentum payoff in month m
$ESG\ MOM_m^{vw}$	Value-weighted ESG momentum payoff in month m
$\hat{\gamma}_{i,1}$	Estimated coefficient on dividend yield used to predict stock returns
$\hat{\gamma}_{i,2}$	Estimated coefficient on yield used to predict stock returns
$\hat{\gamma}_{i,3}$	Estimated coefficient on term spread used to predict stock returns
$\hat{\gamma}_{i,4}$	Estimated coefficient on default spread used to predict stock returns
$\gamma_{i,0}$	Intercept in the stock-level return prediction regression

$\gamma_{i,1}$	Coefficient on dividend yield in the stock-level return prediction regression
$\gamma_{i,2}$	Coefficient on yield in the stock-level return prediction regression
$\gamma_{i,3}$	Coefficient on term spread in the stock-level return prediction regression
$\gamma_{i,4}$	Coefficient on default spread in the stock-level return prediction regression
γ_q	Time dummy variable for quarter q
I_d	Closing level of the index on day d
I_m	Closing level of the index on the last day of month m
$Industry_i$	Industry dummy variable of stock i
INT_m	Intercept in the rolling payoff regression
$Inv_{i,q}$	Total asset growth of stock i in quarter q
K	Number of regressors
λ	Coefficient on the by one period lagged return
$MAX[\cdot]$	Maximum function
$Mom_{i,q}$	Price momentum of stock i in quarter q
N	Total number of stocks in the index
π_0	Intercept in the ESG momentum feedback equation
$P_{i,d}$	Closing price of stock i on day d
$Prof_{i,q}$	Profitability ratio of stock i in quarter q
RES_m	By macroeconomic variables unexplained part of ESG momentum payoffs
ρ	Coefficient linking stock returns to future ESG momentum
$R_{i,q}$	Quarterly return of stock i in quarter q
$R_{i,q-1}$	By one quarter lagged return of stock i
σ_a^2	Variance of the time-invariant individual effect
$\hat{\sigma}_a^2$	Estimated variance of the time-invariant individual effect
σ_u^2	Variance of the idiosyncratic error
$Size_{i,q}$	Natural logarithm of outstanding market capitalization of stock i in quarter q

SSR	Sum of squared residuals from the between regression
T	Number of time periods
$TERM_m$	Term spread in month m
$TERM_{m-2}$	By two months lagged term spread
$TERM_{m-1}$	By one month lagged term spread
θ	Fraction of the time averages subtracted in the random effects transformation
$Value_{i,q}$	Book-to-price ratio of stock i in quarter q
$Var(\cdot)$	Variance
W	Window length in the rolling regression
YLD_m	Annualized 3-month euro area AAA government bond spot rate in month m
YLD_{m-2}	By two months lagged yield
YLD_{m-1}	By one month lagged yield
$\varepsilon_{i,m}$	Error term in the stock-level return prediction regression
$\varepsilon_{i,t}$	Error term in the ESG momentum feedback equation

1 Introduction

The relationship between environmental, social, and corporate governance (ESG) activities and stock returns remains a strongly debated topic in the literature. Researchers have not arrived at a clear consensus on whether firms with stronger ESG characteristics earn higher, lower, or indistinguishable returns relative to other firms with weaker ESG characteristics. Part of this uncertainty arises because ESG activity can be interpreted in several ways. On the one hand, it may improve its public image, labor quality, and profits (see Davis 1973, p. 313). On the other hand, it may proxy for lower risk and therefore lower required returns (see Escobar-Saldivar et al. 2025, p. 9). As a result, the question is not simply "does it pay to be good", as the CSR literature often asks, but rather how ESG-related information is incorporated into stock prices.

In the last two decades, various rating systems have become more broadly available to capture companies' ESG characteristics. These ratings are increasingly used in empirical research to proxy for a firm's ESG activity. Recent studies such as Ye et al. (2022) report a positive relationship between ESG activity and stock returns, whereas Escobar-Saldivar et al. (2025) find a negative association and Kumar (2023) no relevant relationship. Koziol & Roßmann (2022) distinguish the effects of ESG activity across three time periods: before, during, and after firms implement ESG measures. Their described framework implies that ESG activity can generate positive stock returns during the ESG implementation phase without implying permanently higher expected returns after the market incorporates that information into the stock prices. This distinction is important as it suggests that the level of ESG and a change in ESG quality should not be expected to affect returns in the same way.

The differentiation between ESG levels and changes in ESG activity has received more interest in the recent literature through the analysis of ESG momentum, which captures the change in ESG ratings. The concept was introduced by Nagy et al. (2016), who show that firms with recent ESG rating improvements outperformed the MSCI World index over their sample period. A recent study by Escobar-Saldivar et al. (2025) reports for U.S. stocks that high ESG score levels are associated with lower long-term stock returns, while increases in ESG scores are associated with higher returns. Their findings suggest that ESG levels and ESG momentum may proxy different economic mechanisms: the former may reflect lower risk and therefore lower required returns, whereas the latter may capture the repricing that occurs when investors incorporate new ESG information into valuations.

In this thesis, I use pooled OLS, random effects, and fixed effects models to examine whether the sign pattern documented by Escobar-Saldivar et al. (2025) for U.S. stocks also appears in European equity markets between June 2016 and September 2025, specifically among stocks included in the STOXX Europe 600 index. Furthermore, I address potential endogeneity concerns by applying a two-step difference generalized method of moments estimation. Past stock returns may influence subsequent ESG activity and rating changes, which would violate strict exogeneity and bias static panel estimates. The second part of this thesis aims to understand if ESG momentum payoffs reflect firm-specific repricing alone or whether they are partly linked to broader macroeconomic conditions. To do so, I apply the methodological intuition of Chordia & Shivakumar (2002) and investigate whether ESG momentum payoffs vary across business cycle phases observed in the sample period, whether returns predicted by macroeconomic variables such as the market dividend yield, yield, term spread, and default spread differ between high- and low ESG momentum portfolios, whether ESG momentum payoffs change after removing the component of returns predicted by macroeconomic variables, and whether lagged macroeconomic variables explain ESG momentum payoffs directly.

The empirical results suggest that the U.S. sign pattern is also visible among European stocks, although the evidence is weaker. ESG levels seem to be negatively related to stock returns, whereas ESG momentum appears to be positively related to stock returns. The evidence for the European sample is more sensitive to specification choices and sample windows than in the U.S. sample. The business cycle analysis further suggests that ESG momentum payoffs appear to be related to macroeconomic conditions, but the evidence is more supportive than definitive. High ESG momentum portfolios generally exhibit higher returns predicted by macroeconomic variables than low ESG momentum portfolios, and once the predictable component is removed, the remaining payoffs become small and often negative, especially for value-weighted portfolios. Macro variables also seem to explain a meaningful share of the strategy's payoffs directly.

The remainder of this thesis is structured as follows. Chapter 2 reviews the relevant literature, formulates the research questions, and presents the empirical methodology. Chapter 3 describes the data and reports descriptive statistics. Chapter 4 presents the empirical results for the panel regressions and the business cycle analysis and discusses their economic interpretation. Chapter 5 concludes.

2 Literature Review & Methodology

2.1 Literature Review

Corporate Social Responsibility (CSR) has a long history in the academic literature. Bowen (1953) is widely regarded as the foundational contribution to the modern CSR debate (see Carroll 1999, p. 269; Matten & Moon 2008, p. 405; Pollman 2024, p. 409) arguing that the social responsibility of businessmen is *"to pursue those policies, to make those decisions, or to follow those lines of action which are desirable in terms of the objectives and values of our society"*. By contrast, Friedman (1970) says that *"if these are 'social responsibilities', they are the social responsibilities of individuals, not of business"*.

At that stage, the debate was largely conceptual and not empirical. For instance, Davis (1973) sets out both the case for and against corporate social responsibility. On the one hand, he argues that social responsibility may be consistent with long-run profit maximization, because companies operating in better communities benefit from higher-quality labor, lower turnover, lower crime, and an overall better environment for business. Consequently, companies that do not contribute to the accomplishment of social goods may face lower profits in the future. On the other hand, he highlights Friedman's shareholder-oriented view that corporate executives who pursue distinct social causes are spending shareholders' money and are thereby reducing shareholder returns. Hamilton et al. (1993) describe three hypotheses about the relative returns of socially responsible and conventional portfolios. The first one says that there is no difference between socially responsible and conventional portfolio expected returns because social responsibility is not priced, as it is not a proxy for risk. The second hypothesis says that social responsibility is priced so that socially responsible portfolios have lower expected returns since they are less risky. Finally, the third hypothesis states that socially responsible portfolios have higher expected returns than conventional ones due to mispricing.

Early empirical evidence on the performance of socially responsible stocks includes Hamilton et al. (1993), who estimate the excess returns, measured as Jensen's alpha, of socially responsible mutual funds over a benchmark consisting of conventional mutual funds between January 1981 and December 1990. They find no statistically significant excess return, indicating that the market does not price social responsibility characteristics. Statman (2000) estimates the excess return, also measured as Jensen's alpha, of the Domini Social index 400 (DMI), now known as MSCI KLD 400 Social index, over the S&P 500 index between May 1990 and September 1998. The Domini Social index was made up of 400 stocks with strong social characteristics. While he highlights that the

annualized return of the Domini Social index was higher than that of the S&P 500 and the estimated Jensen's alpha was positive, it was not statistically significantly different from zero. He further compared the returns of conventional and socially conscious mutual funds that were on Morningstar's list as of September 1998. Similarly, he did not find a statistically significant difference in performance between the two groups. Using U.S. and international data, Bauer et al. (2005) find that after controlling for size, value, and price momentum, there is no significant difference in returns between ethical and conventional mutual funds between January 1990 and March 2001.

The term ESG refers to Environmental (E), Social (S), and Governance (G) and was introduced in the report "*Who Cares Wins*" published by United Nations (2004), which provided recommendations to strengthen environmental, social, and corporate governance issues in analysis, asset management, and securities brokerage. Since then, various rating systems have become more broadly available to capture companies' ESG characteristics. Providers of those ratings and scores include MSCI, Refinitiv, Sustainalytics, S&P Global, Bloomberg, Moody's, FTSE Russell, ISS, CDP, and KLD¹. More frequently, these ratings are used in empirical research to explore the relationship between ESG activity and stock returns. However, because the evidence remains mixed, the literature has not reached a clear consensus on this relationship.

For instance, Ye et al. (2022) use Bloomberg ESG disclosure scores and show that across 27 EU member states, companies that had higher ESG disclosure in their annual reports also had larger stock returns between 2010 and 2018. In contrast, Ciciretti et al. (2023) show that international firms with high Refinitiv ESG scores had lower expected returns between July 2003 and June 2020. They explain that this negative ESG premium might be driven by lower risk associated with high ESG scores. Similarly, El Ghouli et al. (2011) use an CSR score based on KLD STATS data to examine the effect of corporate social responsibility on the cost of equity for companies in the United States between 1992 and 2007. They find that companies with higher CSR scores have lower cost of equity as they are perceived as less risky. Managers should pursue CSR activities to lower their financing costs and enhance the value of the firm. Escobar-Saldivar et al. (2025) also find that firms with high Refinitiv ESG scores tend to have lower long-term stock returns than companies with lower ESG scores. However, Kumar (2023) reviews 30 papers that used the Fama-French factor model framework to assess the influence of ESG factor exposure on the alpha of stock portfolios and reports mixed empirical evidence rather than a consistent ESG premium in the literature. His review suggests that there is no meaningful relationship between portfolio alpha and ESG factor exposure for globally diversified portfolios, whereas a positive ESG-alpha relationship appears mainly in regional portfolios, particularly those with strong developed-market exposure.

¹See Berg et al. (2022) for a detailed analysis and methodological comparison of popular ratings

Koziol & Roßmann (2022) provide a theoretical explanation for the relationship between ESG activity and expected returns. They distinguish between three time periods: Before awareness of ESG (ante ESG), during implementation of ESG measures (cum ESG), and after ESG implementation (post ESG). Furthermore, they present three market reaction scenarios to firm ESG activity during the three periods. First, investors preferring ESG-compliant firms increase demand for these stocks, reducing the cost of capital. Second, the companies that successfully implement ESG measures are perceived to be more adaptable and innovative, resulting in an increase in expected future cash flows while the cost of capital remains unchanged. Third, ESG activity is perceived as reducing risks, therefore reducing the cost of capital. In all three scenarios, the firm value increases in the cum ESG period, resulting in an increase in returns. However, post ESG, the increased firm value together with the reduced cost of capital suggests a reduced return in scenarios one and three. Only in scenario 2, the increased cash flow results in an unchanged return post ESG compared to ante ESG. They argue that this leads to the conclusion that ESG activity can result in short-term return increases, but does not mean that excess returns can be expected in the long run. Instead, all three scenarios show no long-term increase and possibly declines in expected returns. They further argue that these long-term effects should be unsurprising, since higher expected returns in an information-efficient market require accepting drawbacks such as price risk or illiquidity, and ESG factors cannot be assumed to amplify these disadvantages.

This periodic distinction is useful for interpreting ESG momentum introduced in the empirical literature by Nagy et al. (2016). They propose a strategy based on the 12-month change in a firm's ESG rating. Rather than focusing on the level of ESG ratings, ESG momentum captures how a firm's ESG quality, proxied by an ESG rating, changes over time. They show that a portfolio that overweighted firms whose MSCI ESG rating improved over the previous 12 months outperformed the MSCI World index between February 2008 and March 2015. They argue that this strategy is based on the assumption that future stock performance is related to changes in a firm's ESG quality and that an improvement in ESG ratings signals that the firm is better positioned to mitigate ESG-related risks, resulting in a rapid incorporation into the stock price by the market.

Since then, further evidence has emerged to test the findings of Nagy et al. (2016). For instance, Sverner et al. (2023) demonstrate that among stocks included in the S&P 500 index between June 2014 and June 2021, there exists a statistically significant positive ESG momentum factor premium. They construct the ESG momentum factor using the Fama & French (1993) methodology and ESG scores from Sustainalytics. They also find that rating upgrades had a larger impact on returns than rating downgrades. However, this result lost significance after utilizing the generalized method of moments for

estimation. Chen & Yang (2020) find that investors systematically exaggerate corporate ESG information, leading to the ESG momentum effect. Shanaev & Ghimire (2022) analyze the impact of ESG rating changes on stock returns of US companies between 2016 and 2021 using the calendar-time portfolio methodology. They find that ESG rating upgrades led to positive but inconsistently significant abnormal returns, while downgrades exhibited statistically significant negative risk-adjusted returns. More recent work also emphasizes that markets may price ESG information with lags rather than instantaneously. Specifically, Galema & Gerritsen (2025) find that it takes multiple months for changes in MSCI ESG ratings to be reflected in market prices. They also report that institutional investors changing their holdings around rating changes drives the result. Overall, these findings are contested. Cauthorn et al. (2023) replicated the analysis for US stocks made by Shanaev & Ghimire (2022) and find that rating changes do not significantly affect stock performance in the short-term. They argue that methodological mistakes fully explain the different results.

A recent contribution to the literature made by Escobar-Saldivar et al. (2025) investigates the relationship between ESG scores, ESG momentum, and stock returns for all active, publicly traded stocks with a primary listing in the United States between December 2002 and December 2022. They employ a multi-factor panel data model that contains the 1-year total stock returns as explained variable and ESG scores, changes of ESG scores, and other control variables as explanatory variables. They estimate the model using pooled OLS, random effects, and fixed effects regressions and find that higher ESG score levels are associated with lower stock returns and increases in ESG scores are associated with higher stock returns. They argue that the relationship between stock returns and ESG momentum is driven by signaling theory, meaning that changes in ESG scores are credible signals that reduce information asymmetry and influence investor expectations about future performance. Consequently, they say that improvements in ESG scores can serve as a signal of enhanced corporate quality and reduced non-financial risk, similar to how a bond rating upgrade signals improved creditworthiness.

With regard to the robustness of their results, Escobar-Saldivar et al. (2025) acknowledge the potential endogeneity that would cause their results to be biased and suggest that future research could use the generalized method of moments to address this issue. Endogeneity could arise if stock returns affect future ESG scores or changes in ESG scores exhibiting a dynamic and jointly determined relationship. In fact, Shackleton et al. (2022) document a statistically significant relationship between poor stock market performance and stronger subsequent environmental and social performance. They interpret their findings to mean that a negative stock market performance can motivate firms to improve their social and environmental standing. The potential endogeneity is-

sues are addressed in more detail in the methodology section 2.3.1.2. To the best of my knowledge, prior studies have addressed endogeneity in ESG-return research using the generalized method of moments with a focus mainly on ESG score levels or disclosure-based measures (see e.g. Alsahlawi et al. 2021, p. 13; Naseer et al. 2025, p. 5), whereas direct evidence using ESG momentum appears to be scarce.

Escobar-Saldivar et al. (2025) also highlight that further research is needed to understand the relationship of stock returns and ESG momentum during crisis episodes, such as the Great Recession and the COVID-19 pandemic. There exists literature that has examined ESG scores and ESG momentum during specific periods, like the financial crisis or the COVID-19 pandemic. For example, Lins et al. (2017) show that during the 2008–2009 financial crisis, firms with high pre-crisis Corporate Social Responsibility intensity had larger stock returns than firms with low CSR pre-crisis intensity. Yoo et al. (2021) use daily firm-level Arabesque S-Ray ESG scores during the COVID-19 pandemic and report that increases in ESG, especially the E score, were associated with higher returns during the pandemic. They also find that firms belonging to groups of stocks with lower returns benefited more from increasing their ESG scores than firms that belonged to higher return groups. Magnani et al. (2024) examine ESG momentum among European stocks based on MSCI KLD and Sustainalytics ESG scores in a main sample from July 2009 to December 2019. In an MSCI-only COVID and post-COVID robustness extension covering January 2000 to March 2023, their results become somewhat stronger compared to the pre-COVID period. Both Lins et al. (2017) and Yoo et al. (2021) argue that their results support the argument that higher ESG activity lowers financial risk during financial crises. However, to the best of my knowledge, the literature hasn't examined explicitly how ESG momentum payoffs vary across different market regimes, business cycle phases, or how they are related to macroeconomic variables.

In summary, while the literature on the relationship between ESG score levels and stock returns reports mixed results, theoretical arguments and several empirical studies imply that improvements in ESG characteristics (ESG momentum) may yield positive short-term returns. At the same time, the evidence on ESG momentum has not fully addressed potential endogeneity issues. Furthermore, although ESG characteristics have been examined during specific crisis periods such as the 2008-2009 financial crisis and the COVID-19 pandemic, there is little evidence on how ESG momentum payoffs vary over the business cycle.

2.2 Research Questions

Based on the literature review, this thesis examines three research questions:

- **Q1:** Is ESG momentum positively associated with quarterly stock returns among STOXX Europe 600 stocks?
- **Q2:** Can common macroeconomic variables explain the payoffs to an ESG momentum strategy?
- **Q3:** How are ESG momentum payoffs related to the business cycle?

2.3 Methodology

Section 2.3.1 explains the methodology used to answer research question **Q1**. Section 2.3.2 explains the methodology used to answer research questions **Q2** and **Q3**.

2.3.1 Relation between ESG momentum and Stock Returns among STOXX Europe 600 stocks

To explain quarterly stock returns as a function of ESG scores and ESG momentum, I first estimate the multi-factor model similar to Escobar-Saldivar et al. (2025) shown in Equation 1:

$$R_{i,q} = \beta_0 + \beta_1 Industry_i + \beta_2 Country_i + \beta_3 Beta_{i,q} + \beta_4 Size_{i,q} + \beta_5 Value_{i,q} + \beta_6 Prof_{i,q} + \beta_7 Inv_{i,q} + \beta_8 Mom_{i,q} + \beta_9 ESG_{i,q} + \beta_{10} \Delta ESG_{i,q} + \beta_{11} \gamma_q + u_{i,q} \quad (1)$$

where $R_{i,q}$ denotes the quarterly return of stock i in quarter q . $Beta_{i,q}$ denotes the CAPM beta coefficient, $Size_{i,q}$ the natural logarithm of outstanding market capitalization, $Value_{i,q}$ the book-to-price ratio, $Prof_{i,q}$ the profitability ratio, $Inv_{i,q}$ the total asset growth, $Mom_{i,q}$ the price momentum, $ESG_{i,q}$ the ESG score, and $\Delta ESG_{i,q}$ the 12-month ESG score change of stock i . $Industry_i$, $Country_i$, and γ_q denote industry, country, and time dummy variables, respectively. Each variable is measured on the last day of each quarter. Data sources are provided in the appendix A.3 in table 11.

While the main variables of interest of the analysis in this section are the coefficients $\Delta ESG_{i,q}$ and, closely related, $ESG_{i,q}$, the remaining explanatory variables are included as controls. The following subsections motivate their inclusion in the multi-factor model described by Equation 1 and describe how each control variable is constructed or estimated:

Beta

The foundational Capital Asset Pricing Model (CAPM), introduced in the 1960s by Treynor (1961), Sharpe (1964), Lintner (1965) and Mossin (1966), implies a linear relationship between expected stock returns and market beta. Although CAPM has received limited empirical support, it remains an important theoretical benchmark for subsequent asset pricing research. Since broad market movements affect most stocks regardless of their other characteristics (see Fama & French 1993, p. 5), beta is included in Equation 1 as a firm-level control for exposure to systematic market risk. Based on Bali et al. (2025), for each stock i and each month m , I estimate the rolling beta by running a regression of the excess return of stock i on the excess return of the market portfolio using the past 12 months of monthly data. The market portfolio is proxied by the STOXX Europe 600 index. To obtain $Beta_{i,q}$, I retain the beta estimates from the last month of each quarter. Although Bali et al. (2025) note that beta estimates based on monthly data and only 12 observations are noisier, I decided not to increase the estimation window to retain stocks that do not have a longer return history. To calculate the excess returns, I use the one-month risk-free rate implied by the European Central Bank's (ECB) euro-area AAA government bond yield curve. Specifically, I use the ECB's published Svensson (1994) parameters to calculate the risk-free rate.

Size and Value:

The size effect was discovered by Banz (1981) who, by analyzing NYSE common stocks between 1936 and 1975, finds that small firms, on average, had higher risk-adjusted returns than large firms. Additionally, Reinganum (1981) shows that between 1963 and 1977, small firms systematically outperformed large firms with equivalent beta risk. Early evidence of "value" as a return predictive attribute includes Basu (1977), who finds that during the period from April 1957 to March 1971, portfolios consisting of stocks with low price to earnings ratios have, on average, earned higher absolute and risk-adjusted rates of return than portfolios consisting of stocks with high price to earnings ratios. Furthermore, by analyzing a sample of 1,400 companies between 1980 and 1984, Rosenberg et al. (1985) show that a strategy that buys stocks with a high ratio of book value of common equity per share to market price per share and sells stocks with a low book/price ratio earns, on average, a positive monthly return. Looking at both size and value, Fama & French (1992) show that size and book-to-market equity combine to capture the cross-sectional variation in average stock returns between 1963 and 1990. They argue that if asset-pricing is rational, size and book-to-market must proxy for risk. Fama & French (1993) then introduced the widely used Fama-French three-factor model (FF-3) to measure the returns of size and value, further establishing book-to-price as the benchmark variable to capture value in modern multi-factor literature. However,

later studies such as Horowitz et al. (2000) found that while small firms have achieved higher annualized returns than large firms between 1963 and 1981, this outperformance has vanished during the period between 1982 and 1997. For their analyzed time period, they do not find any support in the data for the use of size as a risk factor. Fama & French (2012) study size and value in international stock returns, including Europe, during the period from November 1990 to March 2011, and find no size premium in any region but a strong value premium. They also note that, except for Japan, value premiums in average stock returns are larger among small stocks. Based on this evidence, value is controlled for in Equation 1 via the regressor $Value_{i,q}$. Although evidence for a distinct size risk factor has been weak since the 1980s, size remains widely used in academic research, including Escobar-Saldivar et al. (2025) and is also a risk factor in the Fama-French five-factor model (FF-5) proposed by Fama & French (2015). It is therefore also controlled for in Equation 1 via the variable $Size_{i,q}$. I calculate $Size_{i,q}$ as the natural logarithm of the outstanding market capitalization of stock i in quarter q . I take the natural logarithm to account for the highly skewed nature of the cross-sectional distribution of market capitalization (see Bali et al. 2025, p. 149). $Value_{i,q}$ is obtained by dividing 1 by the Price-to-Book ratio of stock i in quarter q .

Profitability:

Novy-Marx (2013) analyzed the relation between book-to-market ratio and profitability. He found that companies with higher profitability have higher returns on average. Similarly, companies with higher book-to-market ratios have higher returns. Looking at portfolio characteristics, he also observed a negative correlation between profitability and book-to-market ratio between July 1963 and December 2010, suggesting that the performance of a value strategy can be improved by controlling for profitability and vice versa. Fama & French (2015) measure profitability using operating profitability, while Novy-Marx (2013) relies on gross profitability. Operating profitability is calculated as revenue minus cost of goods sold, minus selling, general, and administrative expenses, minus interest expenses, divided by book equity. Gross profitability, on the other hand, is defined as revenue minus cost of goods sold divided by total assets. Novy-Marx (2013) argue that the “*farther down the income statement one goes, the more polluted profitability measures become, and the less related they are to true economic profitability*” justifying their choice of a simpler profitability measure. Fama & French (2015), however, say that their inferences “*do not seem to be sensitive to the way factors are defined*”. Following Escobar-Saldivar et al. (2025) and Fama & French (2015), I control for profitability in Equation 1 based on operating income and total equity via the variable $Profi_{i,q}$. The profitability ratio $Profi_{i,q}$ is measured as operating income divided by total equity. Due to different reporting frequencies across firms, operating income is not available on a uniform quarterly basis for all companies in the sample. For some com-

panies, only annual or semi-annual operating income is available, while others report quarterly operating income. To ensure a consistent numerator across firms and quarters, I use an annual operating income measure in every quarter. Specifically, I rely on reported annual operating income when available. Otherwise, I calculate annual operating income as the sum of operating income over the trailing four quarters.

Investment:

Titman et al. (2004) document a negative relationship between capital investment and future stock returns. Similarly, Cooper et al. (2008) show that firms with high asset growth earn lower future returns. Fama & French (2015) incorporate this relation into their FF-5 model via the *CMA* factor. Based on this evidence and following Escobar-Saldivar et al. (2025), I control for investment in Equation 1 via the variable $Inv_{i,q}$. I measure investment as the quarterly growth rate in total assets. Specifically, for stock i in quarter q , I calculate $Inv_{i,q}$ as total assets in quarter q divided by total assets in quarter $q - 1$, minus one. For companies that report total assets only annually or semi-annually, I carry the most recently available value forward until updated financial statement data becomes available. Hence, for those months, the quarterly growth rate in total assets will be zero.

Price Momentum:

Price Momentum had appeared in the literature with Levy (1967) who found that stocks which historically were among the 10 percent strongest had a stronger future price appreciation than those stocks that were among the 10 percent weakest. However, this observation was ignored until Jegadeesh & Titman (1993) demonstrated that strategies that buy stocks that have performed well in the past and sell stocks that have performed poorly in the past generate significant positive returns over 3- to 12-month holding periods (see Ang 2014, p. 235). Price Momentum was further established as a factor in asset pricing through the four-factor model introduced by Carhart (1997), who added price momentum as a fourth factor to the Fama & French (1993) three-factor model. In line with Escobar-Saldivar et al. (2025), I control for price momentum in Equation 1 via the variable $Mom_{i,q}$. For each month m and each stock i , I measure price momentum as the 6-month return during the period beginning 7 months prior to and ending one month prior to the measurement date. To obtain $Mom_{i,q}$, I retain the price momentum measure from the last month of each quarter.

Industry, Country and Time Dummies:

I include country and industry dummy variables $Country_i$ and $Industry_i$ to account for systematic differences across firms that arise from their country and industry affiliation.

$Country_i$ represents the country where stock i has its headquarters. This country designation has inherently some limitations as firms operate in multiple countries besides the country in which they have their headquarters. It, therefore, does not capture each political, institutional, or macroeconomic environment that a firm operates in. Nevertheless, it provides a useful proxy for the primary national environment in which the firm is based. Escobar-Saldivar et al. (2025) did not have a country dummy in their multi-factor model as they only analyzed U.S. stocks. $Industry_i$ denotes the Industry Classification Benchmark (ICB) classification of stock i . ICB is an industry classification framework maintained by FTSE Russell, a subsidiary of the London Stock Exchange Group (LSEG), which assigns firms to four hierarchical classification levels (see LSEG 2026, p. 3). $Industry_i$ refers to the highest level classification with 11 possible industries and is broadly comparable to the highest classification level in the GICS framework. Even though GICS is more commonly used in the literature, I use the ICB classification because it was more consistently available for the sample of firms analyzed in this study. Following Wooldridge (2013), I include time dummies to absorb common shocks across time periods.

2.3.1.1 Estimating the multifactor model via Pooled OLS, Fixed Effects, and Random Effects

In this section, I describe the methodology and assumptions of the models used to test the multi-factor model in Equation 1. I first employ the three panel data models, pooled OLS (POLS), fixed effects (FE), and random effects (RE), following Escobar-Saldivar et al. (2025). Each of the three models may be represented by Equation 2:

$$y_{i,t} = \beta' x_{i,t} + a_i + u_{i,t} \quad (2)$$

where $y_{i,t} = R_{i,t}$ denotes the return of stock i in period t , $x_{i,t} \in \mathbb{R}^K$ is the $K \times 1$ vector of regressors, a_i is the time-invariant individual effect of stock i , and $u_{i,t}$ is the idiosyncratic error of stock i in period t . Because a_i is unobserved, it ends up in the error term of the regression. The resulting error $v_{i,t} = a_i + u_{i,t}$ is called the composite error (see Wooldridge 2013, p. 460).

Pooled OLS combines the cross-section and time-series components of the data and estimates the coefficient vector by OLS estimation of $y_{i,t}$ on $x_{i,t}$ (Cameron & Trivedi 2005, p. 702). In order for POLS coefficients to be consistent estimates one has to assume the orthogonality condition $E(x'_{i,t} v_{i,t}) = 0$ and the rank condition $\text{rank}[\sum_{t=1}^T E(x'_{i,t} x_{i,t})] = K$ (see Wooldridge 2001, p. 171). Consistency means that, as the sample size grows, the POLS estimators converge in probability to the true parameters. The rank condition rules out perfect linear dependencies among the explanatory variables. The orthogo-

nality condition requires that the composite error be uncorrelated with the regressors. However, as the composite error contains both the idiosyncratic error and the time-invariant individual effect, the pooled OLS estimates will be biased and inconsistent if the individual effect is correlated with the regressors (see Wooldridge 2013, p. 460). Additionally, pooled OLS does not account for the fact that, because the time-invariant individual effect is included in the composite error, the composite errors will be correlated over time within the same stock.

The random effects model exploits this serial correlation in the error term in a generalized least squares (GLS) framework by building the covariance structure directly into the estimation. For random effects to be consistent, the idiosyncratic error must be strictly exogenous conditional on the full regressor history and the individual effect, $E(u_{i,t}|x_i, a_i) = 0$. Further, the individual effect must be orthogonal to the full regressor history, $E(a_i|x_i) = E(a_i)$, where $x_i = (x_{i,1}, \dots, x_{i,T})$, meaning that the unobserved effect is uncorrelated with the explanatory variables in all periods (see Wooldridge 2001, p. 257). The quasi-demeaning transformation of Equation 2 yields:

$$y_{i,t} - \theta \bar{y}_i = \beta' (x_{i,t} - \theta \bar{x}_i) + (v_{i,t} - \theta \bar{v}_i) \quad (3)$$

with

$$\theta = 1 - [\sigma_u^2 / (\sigma_u^2 + T \sigma_a^2)]^{1/2} \quad (4)$$

where $\sigma_u^2 = \text{Var}(u_{i,t})$, $\sigma_a^2 = \text{Var}(a_i)$, $\bar{y}_i = T^{-1} \sum_{t=1}^T y_{i,t}$, $\bar{x}_i = T^{-1} \sum_{t=1}^T x_{i,t}$, and $\bar{v}_i = T^{-1} \sum_{t=1}^T v_{i,t}$. The random effects transformation subtracts a fraction θ of the time averages from the independent variables, the dependent variable, and the composite error (see Wooldridge 2013, p. 493). Cameron & Trivedi (2005) highlight that a wide variety of estimates for σ_u^2 and σ_a^2 are available. In this study, I use the Python package `Linearmodels` (n.d.) which estimates $\hat{\sigma}_a^2 = \text{MAX}[0, (\frac{SSR}{N-k}) - \frac{\hat{\sigma}_u^2}{T}]$. SSR is the sum of squared residuals from a between regression of $\bar{y}_i$ on an intercept and $\bar{x}_i$. $N - k$ is the degrees of freedom of the between regression. The *max* function ensures that the estimator cannot become negative, in which case the estimation is then by pooled OLS (see Cameron & Trivedi 2005, p. 735). If the random effects assumptions hold, the RE estimator is more efficient than pooled OLS, meaning that it has a smaller asymptotic variance than the POLS estimators. (see Cameron & Trivedi 2005, p. 705; Wooldridge 2001, p. 260). However, if the individual effect is correlated with the regressors, both pooled OLS and random effects will be inconsistent. The fixed effects model accounts for the omitted variable bias by eliminating the individual effect through subtracting the

time averages of each variable in Equation 2 from themselves:

$$y_{i,t} - \bar{y}_i = \beta' (x_{i,t} - \bar{x}_i) + (a_i - \bar{a}_i) + (u_{i,t} - \bar{u}_i) \quad (5)$$

Since the individual effect is constant over time, subtracting the average $\bar{a}_i$ from a_i completely eliminates the unobserved effect. As a result, Equation 5 can be estimated with pooled OLS. However, by subtracting the time averages, one can no longer include any time-constant regressors in x_i (see Wooldridge 2013, p. 485). This effectively eliminates the constant categorical variables *Industry_i* and *Country_i* from Equation 1. Similarly to random effects, the fixed effects model requires that the idiosyncratic error be strictly exogenous conditional on the full regressor history and the individual effect, $E(u_{i,t}|x_i, a_i) = 0$. However, fixed effects does not require that the individual effect be orthogonal to the full regressor history (see Wooldridge 2001, p. 264).

With the advantages and limitations described above of each of the models in mind, following Escobar-Saldivar et al. (2025), I estimate the pooled OLS as the base model, while random effects and fixed effects were considered potentially improved alternatives, evaluated with the Hausman (1978) test. If the null hypothesis of the Hausman test cannot be rejected, the random effects estimates are used rather than fixed effects. Failure to reject either means that the RE and FE estimates are so close that it does not matter which model to use, or the sampling variation of the fixed effects estimates is so large that it cannot be inferred that the differences are statistically significant. If the Hausman test is rejected, the random effects assumption that the unobserved effect a_i is uncorrelated with the explanatory variables is assumed to be false and hence fixed effects is used (see Wooldridge 2013, p. 496). Finally, I report the coefficients for each model alongside entity-clustered standard errors, which are robust to arbitrary serial correlation and heteroskedasticity within panel units, following Wooldridge (2013).

2.3.1.2 Estimating the multi-factor model via Two Step Difference Generalized Method of Moments (GMM)

Although pooled OLS, random effects, and fixed effects estimators offer useful evidence on the association between ESG momentum and stock returns, they potentially do not fully address the econometric challenges of the present analysis. As mentioned before, pooled OLS is vulnerable to omitted-variable bias when unobserved time-invariant firm characteristics are correlated with regressors. Random effects can improve efficiency but requires that the unobserved effect is uncorrelated with the explanatory variables. Fixed effects allows for correlation between the individual effect and the regressors, but still requires strict exogeneity of the idiosyncratic error conditional on the full regressor history.

If, however, stock returns affect future ESG scores and changes in ESG scores, the relationship between ESG momentum and stock returns may be dynamic and jointly determined over time, violating strict exogeneity. A violation of strict exogeneity would result in biased fixed effects estimators (see Wooldridge 2013, p. 486). There exists literature that supports the assumption of endogeneity of ESG momentum. Specifically, Shackleton et al. (2022) document a statistically significant relationship between poor stock market performance – defined as stock returns below the 30th percentile in a given year – and stronger subsequent environmental and social (ES) performance. For each company, they measure ES performance as an industry-adjusted ES index, which is based on five MSCI ESG KLD STATS categories: environment, community, employee relations, diversity, and product. Within each category, the index combines positive ES strengths and negative ES concerns into a net score. They interpret their findings to mean that a negative stock market performance can motivate firms to improve their social and environmental standing. This interpretation is consistent with Lins et al. (2017), who show that during the 2008–2009 financial crisis, firms with high pre-crisis Corporate Social Responsibility (CSR) intensity, had larger stock returns than firms with low CSR pre-crisis intensity. They argue that the trust between a firm and investors, built through investments in social capital, pays off when the overall level of trust in the market suffers a negative shock. Accordingly, they describe investments in social capital as an insurance policy that pays off when the economy at large faces a severe crisis in confidence. Dyck et al. (2019) reinforce this mechanism by arguing that “*a financially motivated investor will ask firms for more E&S after its perceived financial value increases.*”. Together, this motivates the idea that past negative stock performance incentivizes investors to improve ES performance. Shackleton et al. (2022) further show that the negative relationship between current stock returns and future ES performance tends to be stronger among firms facing fewer financial constraints – such as those with lower leverage ratios, higher free cash flow, and greater equity repurchases – consistent with the view that cash-abundant firms can afford to invest in CSR-improving activities.

If poor stock market performance truly impacts future ESG-related activities, then ESG scores and ESG momentum cannot be treated as strictly exogenous in equation 1. The violation of this feedback mechanism on the strict exogeneity assumption can be shown using Wooldridge (2001), who expresses strict exogeneity as $E(y_{i,t}|x_{i,1}, \dots, x_{i,T}, a_i) = E(y_{i,t}|x_{i,t}, a_i)$. It requires that once x_{it} and a_i are controlled for, the explanatory variables of other time periods, including future ones, must have no partial effect on $y_{i,t}$. This is the equivalent of $E(u_{i,t}|x_i, a_i) = 0$. Suppose now that ESG momentum in t depends on the stock return in $t - 1$:

$$\Delta ESG_{i,t} = \pi_0 + \rho y_{i,t-1} + \dots + \varepsilon_{i,t} \quad (6)$$

where $\rho \neq 0$. If you insert Equation 2 into the variable $y_{i,t-1}$ in Equation 6, you get:

$$\Delta ESG_{i,t} = \pi_0 + \rho(\beta' x_{i,t-1} + a_i + u_{i,t-1}) + \dots + \varepsilon_{i,t} \quad (7)$$

Consequently, ESG momentum in period t would contain the return shock $u_{i,t-1}$, violating strict exogeneity. To account for endogeneity, I additionally estimate a dynamic multi-factor model shown in Equation 8 using two-step difference generalized method of moments (GMM), which uses lagged variables as internal instruments and is designed for panel data with fixed effects and regressors that are not strictly exogenous:

$$R_{i,q} = \lambda R_{i,q-1} + \beta_3 Beta_{i,q} + \beta_4 Size_{i,q} + \beta_5 Value_{i,q} + \beta_6 Prof_{i,q} + \beta_7 Inv_{i,q} + \beta_8 Mom_{i,q} + \beta_9 ESG_{i,q} + \beta_{10} \Delta ESG_{i,q} + \beta_{11} \gamma_t + u_{i,q} \quad (8)$$

The dynamic multi-factor model in Equation 8 may be represented by Equation 9:

$$y_{i,t} = \lambda y_{i,t-1} + \beta' x_{i,t} + a_i + u_{i,t} \quad (9)$$

where $y_{i,t}$ denotes the return of stock i in period t , $y_{i,t-1}$ denotes the by one period lagged return, $x_{i,t} \in \mathbb{R}^K$ is the $K \times 1$ vector of regressors, a_i is the time-invariant individual effect, and $u_{i,t}$ is the idiosyncratic error in period t . As explained by Roodman (2009) and Arellano & Bover (1995), two-step difference GMM first removes the individual effect a_i using forward orthogonal deviations. It does so by subtracting the average of all future available observations of a variable. This is advantageous over first-differencing, which subtracts previous observations from the contemporaneous ones, as it minimizes data loss in unbalanced panels. Since lagged observations do not enter the transformation, they are valid instruments. The transformed dynamic multi-factor model and the forward orthogonal deviations adjusted explanatory variables are:

$$y_{i,t}^* = \lambda y_{i,t-1}^* + \beta' x_{i,t}^* + u_{i,t}^* \quad (10)$$

$$x_{i,t}^* = c_t \left(x_{i,t} - \frac{1}{(T-t)} \sum_{s=t+1}^T x_{i,s} \right) \quad (11)$$

and the scale factor $c_t = \sqrt{(T-t)/(T-t+1)}$. Just as for fixed effects, removing the unobserved effect does not solve the endogeneity issue described before. Arellano & Bond (1991) show that, under the assumption that the idiosyncratic errors are not seri-

ally correlated, sufficiently old lagged levels of the explanatory variables can be used as valid instruments for the transformed variables. Next, Roodman (2009) explains that once the model has been transformed and instrumented, GMM performs an initial regression, obtains the residuals from this estimation, and uses the residuals to construct a sandwich proxy for the covariance matrix of the moment conditions. The GMM regression is then rerun using the proxy for the covariance matrix. The results of this two-step approach yield efficient and robust coefficients. As highlighted by Roodman (2009), because the clustered covariance matrix only assumes that errors are correlated within firms, I use time dummies to remove universal time-related shocks from the errors.

Escobar-Saldivar et al. (2025) acknowledge the potential endogeneity, but argued that due to the large time dimension in their panel of 81 quarterly observations, pooled OLS, random effects, and fixed effects models remained consistent and efficient. Wooldridge (2001) says that under bounded moments and weak dependence, the fixed-effects bias declines at rate $1/T$, so with large T the bias may be minimal. Likewise, Roodman (2009) explains that with a large time dimension, dynamic panel bias in the within estimator becomes insignificant, and fixed-effects works. Meanwhile, in GMM estimation, the number of instruments increases rapidly with a larger time dimension, which may overfit endogenous variables, leading to biased coefficients unless the instrument count is restricted. Since the time dimension of this study contains only 38 quarterly observations, the panel is not clearly in a "large- T " setting as Escobar-Saldivar et al. (2025) may have been. Nevertheless, as this is a valid concern, for robustness, I re-estimate the GMM model over shorter sub-periods. Roodman (2009) notes that there is no general rule for how many instruments are too many, but that the `xtabond2` command in the software Stata warns when instruments outnumber individual units in the panel, as a cautious rule of thumb. To limit the instrument count, I collapse the instrument matrix as suggested by Roodman (2009). Wintoki et al. (2012) state that using internal instruments eliminates the need for external instruments, which might be difficult to obtain. However, they also explain that *"there is a potential problem with weak instruments, which becomes greater as the number of lags of the instrumental variables increases"*.

After performing the generalized method of moments model, I apply two post-estimation tests to determine that an appropriate econometric model is applied. These are the Hansen test of overidentifying restrictions and the Arellano-Bond test for first-order (AR (1)) and second-order (AR (2)) correlation. Roodman (2009) explains that the Hansen test evaluates the joint validity of the overidentifying restrictions, meaning it checks whether the full instrument set is collectively consistent with the assumption of exogeneity. Its null hypothesis is that the instruments are valid, so that a high p-value implies that one cannot reject instrument validity. However, Roodman (2009) warns that the Hansen test should not be relied upon too faithfully as it can be weakened if there

are too many instruments producing implausibly good p-values close to 1.000. He suggests to view p-values above 0.25 as a potential sign of a weak Hansen test. A p-value below 0.1 would potentially indicate invalid instruments. The Arellano-Bond tests are applied to the residuals in first differences and are used to detect serial correlation in the idiosyncratic error term, which could invalidate some lags as instruments. Because $\Delta u_{i,t}$ is mechanically related to $\Delta u_{i,t-1}$ as they both contain $u_{i,t-1}$, negative first-order serial correlation in differences is expected and therefore uninformative. For that reason, second-order correlation in differences is used to detect first-order serial correlation in the level errors. An insignificant AR (2) p-value indicates that there is no evidence of second-order serial correlation, which means that the validity of the chosen lagged levels as instruments cannot be rejected.

Finally, I will report the coefficients obtained by performing the above-explained GMM estimation alongside Windmeijer (2005) corrected standard errors next to the pooled OLS, random effects, and fixed effects results for comparison.

2.3.2 Business Cycle Analysis

This part of the thesis focuses on the relationship between ESG momentum and the business cycle. Following Chordia & Shivakumar (2002), I capture different market phases using the macroeconomic variables that prior studies have shown to predict returns. Specifically, I predict the return r of stock i in month m by the following Equation:

$$r_{i,m} = \gamma_{i,0} + \gamma_{i,1}DIV_{m-1} + \gamma_{i,2}YLD_{m-1} + \gamma_{i,3}TERM_{m-1} + \gamma_{i,4}DEF_{m-1} + \varepsilon_{i,m} \quad (12)$$

where $r_{i,m}$ denotes the monthly return of stock i in month m , DIV_{m-1} denotes the by one month lagged dividend yield on the market, YLD_{m-1} denotes the lagged 3-month AAA government bond spot rate, $TERM_{m-1}$ denotes the lagged term spread, and DEF_{m-1} denotes the lagged default spread. The inclusion of these explanatory variables in Equation 12 is motivated as follows:

Dividend Yield (DIV):

The dividend yield on the market DIV_{m-1} is defined as the aggregate dividend payments of the firms included in the STOXX Europe 600 over the previous 12 months divided by the current level of the index. There is extensive research that shows that dividend yields are associated with returns. Rozeff (1984) finds that stock market returns increase as the market dividend yield in the prior year increases. Campbell & Shiller (1988), using broad U.S. stock indexes, show that stock returns are somewhat predictable and that the lagged log dividend-price ratio has a positive effect on stock returns. Fama & French

(1988) use dividend-price ratios to predict stock returns and find that forecast power increases with time horizon, and argue that stock prices are low relative to dividends when expected returns are high, so that a high dividend yield predicts high future returns. For non-U.S. research, Wetherilt & Wells (2004) report U.K. evidence that dividend yield predicts long-horizon excess returns, although this predictive ability strongly depends on the sample period. Monteiro et al. (2020) find the strongest supportive evidence of return predictability for the U.S., the U.K., and France, while Germany shows some evidence of short-term return predictability; by contrast, Spain, Italy, and Japan exhibit weak or no clear return predictability. Based on this, although mixed, U.S. and European evidence, and following the approach by Chordia & Shivakumar (2002), I include the dividend yield of the STOXX Europe 600 index as a proxy for the time variation in the unobservable risk premium. A high dividend yield serves as indicator that dividends are being discounted at a higher rate. Da et al. (2014) define the index level dividend-to-price ratio as the value-weighted average of firm-level dividend-to-price ratios. Adapting this logic to the STOXX Europe 600 index, the impact of dividend payments, expressed as dividend points, of stock i on ex-dividend date d can be expressed as its dividend-to-price ratio $\frac{D_{i,d}}{P_{i,d}}$ multiplied by its index weight $w_{i,d}$ and the index level I_d . The trailing-twelve-month dividend points at month-end m are therefore obtained by summing these constituent-level contributions over all ex-dividend dates within the preceding twelve months:

$$DP_m^{TTM} = \sum_{i=1}^N \sum_d^{EXD} \left(\frac{D_{i,d}}{P_{i,d}} \right) \cdot w_{i,d} \cdot I_d \quad (13)$$

where DP_m^{TTM} denotes the sum of dividend points on the last day in month m over the period $m - 11$ to m of the STOXX Europe 600 index, $D_{i,d}$ denotes the dividend payment of stock i on day d , $P_{i,d}$ denotes the closing price of stock i on day d , $w_{i,d}$ denotes the relative free-float market capitalization of stock i on day d capped at 20% (STOXX 2025, p. 5), I_d denotes the closing level of the index on day d , N denotes the total number of stocks in the index, and EXD denotes all ex-dividend dates over the period $m - 11$ to m . I include the dividend payments with payment types of cash dividend, dividend with stock alternative, and stock dividend with cash alternative. Following FTSE (n.d.), I assume that investors will always elect cash payments over stock alternative options and therefore treat all three payment types as cash dividends. I exclude sole stock dividends. This classification of payment types is based on Refinitiv payment-type data. The dividend yield of the STOXX Europe 600 index in month m is then calculated as:

$$DIV_m = \frac{DP_m^{TTM}}{I_m} \quad (14)$$

where DIV_m is the dividend yield on the last day of month m , DP^{TTM} is the sum of dividend points over the previous 12 months of Equation 13, and I_m denotes the closing level of the index on the last day of month m .

Yield (YLD):

The yield YLD_{m-1} is defined as the annualized 3-month euro area AAA government bond spot rate published by the European Central Bank (ECB). Fama & Schwert (1977) use the nominal return on a three-month U.S. treasury bill as proxy for expected quarterly inflation rate and show that common stock returns show negative relationships with the quarterly expected inflation proxy. Similarly, Fama (1981) finds a negative relation between stock returns and expected inflation rates proxied by the three-month Treasury bill in the quarterly specification. Ang & Bekaert (2007) investigate the relationship between stock returns and the 1-month EURO rate from Datastream as the short rate. At short horizons, they find that the short rate strongly negatively predicts returns and that these results are robust in international data. Hjalmarsson (2010) also reports that local short interest rate measures of developed countries are fairly robust predictors of excess stock returns. Those short interest rate measures include the rates on local 3-month T-bills when available or, otherwise, private discount rates or interbank rates. Charles et al. (2017) further support these findings by presenting that the short-term interest rate, defined as local 3-month Treasury bill rate when available and money market rate or 3-month interbank rate otherwise, is a strong predictor for stock returns, specifically for European markets. While I could not identify a paper that uses the exact ECB series used in this study, following the approach of Chordia & Shivakumar (2002), I include it to serve as a European short-rate proxy for expectations of future economic activity since the international literature consistently treats the local short rate - typically a 3-month treasury bill rate, money market rate or interbank rate - as the relevant predictor of future stock returns, and because using the local short rate is the international equivalent to the standard U.S. forecasting regression. The ECB series is therefore a defensible euro-area counterpart, as it measures a euro-denominated, annualized 3-month AAA government spot rate, which is a high-credit-quality short-horizon rate.

Term Spread (TERM):

The term spread $TERM_{m-1}$ is defined as the difference between the annualized 10-year euro area AAA government bond spot rate and the annualized 3-month euro area AAA government bond spot rate published by the European Central Bank. Fama & French (1989) measure the term spread as the difference between the Aaa yield and the one-month bill rate and show that the term spread is related to the shorter-term business cycles identified by the NBER. Bonser-Neal & Morley (1997) evaluated 11 industrial countries and found that the term spread is a statistically and economically significant

predictor of real economic activity in several industrial countries besides the United States, including Germany, France, the Netherlands, and Canada. To ensure comparability with the U.S. literature, they define, whenever possible, the term spread as the difference between the local rate on the 10-year government bond and the local 3-month government bill rate. In countries where these assets are not actively traded, they use the closest actively traded long-term government bond or the rate on interbank deposits. Chordia & Shivakumar (2002) measure the term spread as the difference between the average yield of Treasury bonds with more than 10 years to maturity and the average yield of T-bills that mature in three months. Based on this evidence, and given the cross-country European setting of this study, I include the annualized spread between 10-year and 3-month euro area AAA government bond spot rates as term spread in Equation 12. This choice is also supported by the ECB's characterization of the AAA government bond curve as risk-free and consistent reference term structure to market participants on a daily basis (see ECB 2008, p. 95). According to ECB (2008), zero-coupon yields are not directly observable for a wide range of maturities. Therefore, both the 3-month and the 10-year euro area AAA government bond spot rates used here are not observed yields of individual bonds but model-implied zero-coupon yields obtained by fitting a Nelson-Siegel-Svensson model yield curve to the yields of eligible AAA euro area government bonds and then evaluating the fitted curve at 3-month and 10-year maturities, respectively. Svensson (1994) provides the original, while ECB (n.d.) and ECB (2008) describe the ECB's implementation in detail. It should be mentioned that not every country in this study is part of the euro area. For those countries, the 3-month and 10-year yields serve as proxies for the local rate.

Default Spread (DEF):

The default spread DEF_{m-1} is defined as the difference between the average yield of bonds rated BAA by Moody's and the average yield of bonds with a Moody's rating of AAA, and is included to capture the effect of default premiums. Fama & French (1989) define a related default spread as the difference between the yield on a market portfolio of corporate bonds and the yield on Aaa bonds. They report that the default spread is a business-conditions variable, high during recessions and low when the economy is persistently strong. Bundesbank (2022) states that Moody's Baa–Aaa spread is a well-known indicator of global risk. De Santis (2012) further supports this, who describes that the US Baa–Aaa corporate bond spread is generally used to measure global uncertainty. Following Chordia & Shivakumar (2002), who include the default spread regressor in their model, I also include the Moody's Baa–Aaa spread. Although the Moody's Baa–Aaa spread is based on U.S. corporate bonds, given the aforementioned evidence and due to the absence of a comparable and consistently available Europe-wide series, I interpret it as a proxy for global business cycle conditions.

2.3.2.1 Forming ESG Momentum Portfolios and Calculating ESG Momentum Payoffs

This section explains how I form portfolios based on ESG momentum and how I calculate ESG momentum payoffs. Those portfolios and payoffs are used for the analysis in sections 2.3.2.2 through 2.3.2.5. Each month $m - 1$, the stocks currently present in the STOXX Europe 600 index are sorted based on their 12-month ESG score change (ESG momentum) and assigned to one of five portfolios based on their ranking. The top 20% of stocks with the highest ESG momentum are added to portfolio five (P5) and the bottom 20% of stocks with the lowest ESG momentum are added to portfolio one (P1). The ESG momentum strategy takes a long position in portfolio P5 and a short position in portfolio P1. The positions are held for the following month m , referred to as the holding period. According to Fisher & Lorie (1970), a diversified portfolio requires 32 stocks to achieve a 95% reduction in firm-specific return variability, while 128 stocks are needed for a 99% reduction. Similarly, Campbell et al. (2001) conclude that at least 50 stocks are necessary to construct a diversified portfolio. In light of these findings, I decided to form no more than five portfolios to ensure a sufficient portfolio size each month. Between June 2016 and July 2025, the number of stocks in each portfolio ranged from 106 to 119.

$ESG_MOM_m^{ew}$ denotes the equal-weighted ESG momentum payoff in month m , that is, the return on the ESG momentum strategy in which each stock within a portfolio receives the same weight. $ESG_MOM_m^{vw}$ denotes the value-weighted ESG momentum payoff in month m , that is, the ESG momentum strategy return in which stocks within each portfolio are weighted by their outstanding market capitalization obtained in $m - 1$. Hou et al. (2017) replicated 452 anomalies published in the finance literature and showed that microcap stocks accounted for many of them. Novy-Marx & Velikov (2016) assert that equal-weighted portfolios are more popular in the academic literature because they are often stronger than their value-weighted counterparts. Consequently, equal-weighted portfolios should be viewed skeptically. Further, they argue that small stocks are more expensive to trade as they are not as liquid, which often results in a deterioration of net performance. By construction, the equal-weighted strategy takes disproportionately large positions in small stocks. As a result, the payoffs from that strategy would possibly not be attainable by an investor, as he faces higher trading costs. For robustness, I therefore perform the next-described analysis for both equal-weighted and value-weighted portfolios where applicable.

2.3.2.2 Business Cycle Analysis 1: ESG Momentum Payoffs Over Time Cycles

In this section, I start by calculating the cumulative ESG momentum payoffs $ESG_MOM_m^{ew}$ and $ESG_MOM_m^{vw}$ over time and comparing those returns with the returns of other fac-

tors, obtained from the Fama-French Data Library². Then, following Chordia & Shivakumar (2002), I calculate the ESG momentum payoffs in the holding periods that are classified by recession and non-recession periods as determined by the CEPR–EABCN Euro Area Business Cycle dates.³ If there is a difference between recession and non-recession periods, this could suggest that ESG momentum might be related to the business cycle. This serves only as a base analysis as there is only one short recession phase (COVID-19 pandemic) included in the here analyzed sample period.

For all tests based on monthly time-series averages and rolling time-series regressions in sections 2.3.2.2 through 2.3.2.5, I report heteroskedasticity and autocorrelation consistent (HAC) Newey & West (1987) standard errors. The lag length is chosen as the integer part of $g = [4(n/100)^{2/9}]$, where n is the number of monthly observations used in the relevant time-series regression or average. The integer g controls how much serial correlation is allowed in computing the standard errors (see Wooldridge 2013, p. 432).

2.3.2.3 Business Cycle Analysis 2: Predicted Returns Across ESG Momentum Portfolios

Following the intuition outlined by Chordia & Shivakumar (2002), I examine whether the component of stock returns that is predictable from macroeconomic variables differs across ESG momentum portfolios. If stocks in the portfolio with the highest ESG score upgrades (P5) have statistically significantly different by macro variables predicted returns than the stocks in the portfolio with the largest ESG score downgrades (P1), then this would suggest that the component of returns related to macroeconomic variables varies systematically across ESG momentum portfolios and that differences in predicted returns across ESG momentum portfolios may account for the strategy’s payoffs.

For this analysis, I estimate the rolling regression in Equation 12 separately for each stock i . For robustness, I run the regression using the past 36, 48, and 60 months of data. Specifically, for a given window length $W \in \{36, 48, 60\}$, I estimate the model using monthly observations from months $m - W$ through $m - 1$. Thus, neither the formation month m nor the subsequent holding month $m + 1$ is included in the estimation window. I then compute the predicted returns for stock i in month m and in month $m + 1$ using the coefficients estimated from the regression.

I repeat this estimation for each formation month in the sample. As a result, for each stock and each formation month, two predicted returns are obtained: one for event month 0 (the formation month) and one for event month 1 (the holding month). Next, stocks are assigned to ESG momentum portfolios in formation month m . They stay in

²See https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

³See <https://eabcn.org/dbc/graphs>

the same portfolio during holding month $m + 1$. I then calculate, separately for event month 0 and event month 1, the pooled median predicted return of the stocks in portfolios P1 and P5, where the pooling is done across all stock-level predicted returns from all formation months within a given portfolio and event month. The comparison between P1 and P5 shows whether stocks with low and high ESG momentum differ in their macro-predicted returns at formation and during the subsequent holding month. To determine statistical significance, for each event month and each month in the sample, I compute the difference between the median predicted returns of P5 and P1 and apply a nonparametric two-sided sign test to assess whether positive and negative differences between the two portfolios are equally likely. Additionally, I apply a nonparametric one-sided greater sign test to test whether positive differences between the two portfolios are more likely than negative differences. I use a sign test here as the focus is examining directional consistency of the difference, and a sign test does not require parametric distributional assumptions such as normality for the paired differences (see Fay 2022, p. 84; Nahm 2016, p. 8). A significant result in the one-sided greater sign test would indicate that stocks in P5 tend to have higher macro-predicted returns than stocks in P1.

2.3.2.4 Business Cycle Analysis 3: ESG Momentum Payoffs After Adjusting for Predicted Returns

Following the intuition of Chordia & Shivakumar (2002), I analyze if there are differences in ESG momentum payoffs after removing the by macroeconomic variables predicted part of returns. If the remaining components of returns are smaller and insignificant, this would suggest that ESG score changes do not predict the portion of future returns that are unexplained by the business cycle model, and the predictive ability of ESG score changes is restricted to the portion of returns that is predictable by macroeconomic variables. I estimate the rolling regression in Equation 12 separately for each stock i . For robustness, I run the regression using the past 36, 48, and 60 months of data. Specifically, for a given window length $W \in \{36, 48, 60\}$, I estimate the model using observations from months $m - W$ through $m - 1$. At least 24 months of observations within the window are required for training. The coefficient estimates are used to predict the one-month-ahead stock return in month m based on the macro variables observed in $m - 1$. To isolate the component of returns attributable to time-varying macroeconomic variables, I exclude the estimated intercept from the predicted return:

$$\hat{r}_{i,m} = \hat{\gamma}_{i,1}DIV_{m-1} + \hat{\gamma}_{i,2}YLD_{m-1} + \hat{\gamma}_{i,3}TERM_{m-1} + \hat{\gamma}_{i,4}DEF_{m-1} \quad (15)$$

Next, for each month m , I calculate the adjusted returns as the difference between the realized return $r_{i,m}$ and the predicted return $\hat{r}_{i,m}$. They represent the by macroeconomic

variables unexplained part of returns: $r_{i,m}^{ADJ} = r_{i,m} - \hat{r}_{i,m}$. For robustness, I also calculate the adjusted returns based on the predicted return that includes the estimated intercept. For each holding month, I aggregate the stock-level adjusted returns within each ESG momentum portfolio to obtain equal-weighted and value-weighted portfolio-level adjusted returns. The ESG momentum adjusted payoff $ESG_MOM_ADJ_m^{ew/vw}$ is then defined as the difference between the adjusted returns of portfolio P5 and portfolio P1 in the holding month. I report the adjusted payoffs and the percentage of ESG momentum payoffs that are positive. For comparison, I also compute the corresponding raw ESG momentum payoff over the same set of holding months.

2.3.2.5 Business Cycle Analysis 4: ESG Momentum Payoffs Regressed on Macroeconomic Variables

Following the methodology outlined by Chordia & Shivakumar (2002), I investigate if the macroeconomic variables directly explain the monthly ESG momentum payoffs $ESG_MOM_m^{ew}$ and $ESG_MOM_m^{vw}$. If the macro conditions fully drive ESG momentum payoffs, the average unexplained payoff should be zero. For each holding month $m - 1$, I run a rolling regression of the payoffs $ESG_MOM_{m-1}^{ew}$ and $ESG_MOM_{m-1}^{vw}$ on the macro variables DIV , YLD , $TERM$, and DEF obtained in $m - 2$. For robustness, I run the regression using the past 36, 48, and 60 months of data. The betas and intercept in regression 16 are labeled with the subscript m as those are used for predicting payoffs in month m in the next step, even though they are obtained from the rolling regression estimated on the historical window ending in $m - 1$:

$$ESG_MOM_{m-1}^{ew/vw} = INT_m + \beta_{DIV,m}DIV_{m-2} + \beta_{YLD,m}YLD_{m-2} + \beta_{TERM,m}TERM_{m-2} + \beta_{DEF,m}DEF_{m-2} + u_{m-1} \quad (16)$$

I then use the estimated coefficients from regression 16 to predict the payoffs $ESG_MOM_m^{ew}$ and $ESG_MOM_m^{vw}$ using the macroeconomic variables obtained in $m - 1$. The intercept from regression 16 is not included in order to only capture the by macroeconomic variables predicted ESG momentum payoff:

$$\widehat{ESG_MOM}_m^{ew/vw} = \hat{\beta}_{DIV,m}DIV_{m-1} + \hat{\beta}_{YLD,m}YLD_{m-1} + \hat{\beta}_{TERM,m}TERM_{m-1} + \hat{\beta}_{DEF,m}DEF_{m-1} \quad (17)$$

Next, for each month m , I calculate the by macroeconomic variables unexplained part RES_m as the realized payoff $ESG_MOM_m^{ew/vw}$ minus the predicted payoff $\widehat{ESG_MOM}_m^{ew/vw}$. Lastly, I report the average unexplained payoff RES , the percentage of ESG momentum payoffs that are positive, and the average coefficients from regression 16.

3 Data

This study is based on European individual stock data obtained from Refinitiv covering the period from June 2016 to September 2025. The analysis described in section 2.3.1 uses quarterly data since financial statement variables such as operating income were not available at a more granular frequency. The analysis described in section 2.3.2 uses monthly data. The stock universe contains all stocks that were constituents of the STOXX Europe 600 index. The STOXX Europe 600 is a free-float market capitalization-weighted index that launched in June 1998. It contains the 600 largest stocks from 17 developed European countries¹ representing nearly 90% of the underlying investable market (see STOXX 2025, p. 3). To avoid survivorship bias, each quarterly and monthly sample includes only those stocks that were constituents of the STOXX Europe 600 index in the respective quarter or month. Since I could not obtain constituent lists for the STOXX Europe 600 from Refinitiv prior to June 2016, the analysis in this study begins in that month. For balance sheet variables, observations are dated according to the official release date of the company's financial results. Dividend payments are dated to the dividend ex date. The ESG scores are also obtained from Refinitiv. After the London Stock Exchange Group (LSEG) acquired Refinitiv in 2021, the Refinitiv ESG score was rebranded as LSEG ESG score. However, in this study, I refer to it as Refinitiv ESG score. The Refinitiv ESG database covers over 90% of the global market cap with a history dating back to 2002. LSEG calculates over 870 company-level ESG measures, which are grouped into 10 categories that are aggregated into three pillar scores: Environmental, Social, and Governance. The overall ESG score is a weighted company score based on these pillar scores. The weights vary per industry for the environmental and social categories. For governance, the weights remain the same across all industries. Following corporate reporting cycles, the database is continuously updated, with the data being adjusted weekly and the ESG scores being recalculated accordingly. However, since companies publish their ESG reports annually, the scores remain unchanged throughout the fiscal year of each company (see LSEG 2024, p. 5). Although Denmark, Norway, Poland, Sweden, Switzerland, and the United Kingdom are outside the euro area, all returns and accounting variables are measured in euros to reflect a euro-investor perspective. Exchange rate risk is therefore not modeled explicitly and may affect the results (cf. Fama & French 2012, p. 459).

Table 1 provides summary statistics for the unwinsorized variables used in this study. Over the entire sample period, the average stock has a mean quarterly return of 1.81%, a CAPM beta coefficient of 1.082, a book-to-price ratio of 57.23%, a profitability ratio of 22.13%, a quarter-over-quarter asset growth of 2.65%, a 6-month stock price change

¹These 17 countries are Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Italy, Luxembourg, the Netherlands, Norway, Poland, Portugal, Spain, Sweden, Switzerland, and the United Kingdom.

Table 1: Summary Statistics

This table presents summary statistics for the period from June 2016 to September 2025. *Beta, Size, Value, Profitability, Investment, Price Momentum, ESG Score, ESG Momentum, DIV, YLD, TERM, and DEF* were described in chapter 2.3. Stock Return (Q) is the quarterly stock return, Stock Return (M) is the monthly stock return, and MCap is the outstanding market capitalization. For the cross sectional variables *Stock Return (Q), Stock Return (M), Beta, Size, MCap, Value, Profitability, Investment, Price Momentum, ESG, and ESG Momentum*, each quarter, the mean (*Mean*), standard deviation (*SD*), skewness (*Skew*), kurtosis (*Kurt*), minimum (*Min*), 0.5th percentile (*0.5%*), fifth percentile (*5%*), 25th percentile (*25%*), median (*Median*), 75th percentile (*75%*), 95th percentile (*95%*), 99.5th percentile (*99.5%*), and maximum (*Max*) of the cross-sectional distribution of each variable are calculated. This table presents the time-series means for each cross-sectional value. For the monthly variables *Dividend Yield, Yield, Term Spread, Default Spread*, the same parameters are calculated as simple time series statistics. Column n indicates the average number of stocks for which the given variable is available.

	Mean	SD	Skew	Kurt	Min	0.5%	5%	25%	Median	75%	95%	99.5%	Max	n
Stock Return (Q)	0.0181	0.1310	0.5277	4.0423	-0.4620	-0.3154	-0.1756	-0.0638	0.0129	0.0916	0.2283	0.4458	0.7207	531
Stock Return (M)	0.0049	0.0746	0.1594	5.2080	-0.3401	-0.2066	-0.1086	-0.0386	0.0039	0.0472	0.1212	0.2342	0.3806	585
MCap	19.8526	32.3949	4.3363	24.8714	1.4137	1.8135	2.5200	4.7631	8.7415	20.3769	75.1011	210.2260	299.1184	585
Beta	1.082	0.7732	0.3358	1.2666	-1.5666	-0.8477	-0.0731	0.5769	1.041	1.5508	2.3669	3.4630	4.3414	531
Size	23.0352	1.0373	0.7471	0.1477	21.1523	21.3272	21.6458	22.2655	22.8499	23.6762	25.0558	26.0845	26.3947	531
Value	0.5723	0.6090	4.5645	42.2715	-0.6976	-0.1424	0.0794	0.2183	0.4106	0.7475	1.5185	3.4577	7.3888	531
Profitability	0.2213	0.8436	2.9082	181.6107	-8.2635	-1.0750	-0.0503	0.1102	0.1868	0.2823	0.5680	1.9979	12.607	531
Investment	0.0265	0.1859	9.0222	147.4305	-0.3644	-0.1607	-0.0434	-0.0022	0.0033	0.0242	0.1175	0.7030	3.2664	531
Price Momentum	0.0372	0.1893	0.8696	3.9597	-0.5395	-0.4120	-0.232	-0.0832	0.0244	0.1366	0.3488	0.7161	1.0789	531
ESG Score	67.2263	15.4759	-0.7881	0.4905	8.7821	19.9738	37.7607	58.0424	69.6416	78.8027	88.2009	92.8206	94.2944	531
ESG Momentum	1.5798	5.5709	0.9780	6.0434	-20.467	-11.743	-5.8068	-1.6236	0.9353	4.1300	11.1692	22.0315	34.6089	531
Dividend Yield	0.0351	0.0045	0.0524	-0.3544	0.0259	0.0259	0.0271	0.0329	0.0350	0.0377	0.0432	0.0459	0.0459	
Yield	0.0041	0.0079	-0.7171	-0.1220	-0.0170	-0.0170	-0.0110	0.0010	0.0054	0.0107	0.0136	0.0192	0.0192	
Term Spread	0.0087	0.0020	0.9470	1.2378	0.0060	0.0060	0.0064	0.0070	0.0085	0.0103	0.0117	0.0170	0.0170	
Default Spread	0.0048	0.0169	0.9299	-0.8774	-0.0093	-0.0093	-0.0080	-0.0070	-0.0062	0.0194	0.0378	0.0382	0.0382	

(price momentum) of 3.72%, an ESG score of 67.23, a 12-month ESG score change (ESG momentum) of 1.58, and a mean monthly market capitalization of 19.85 billion €. The high average levels of skewness and kurtosis, especially of the Value, Profitability, and Investment variables, indicate that the variables' distributions are heavily influenced by outliers. To mitigate the heavy tails, for the panel regressions, all explanatory variables are winsorized at the 0.5% and 99.5% level on a quarterly basis, following Bali et al. (2025). Although the STOXX Europe 600 index comprises 600 stocks, missing values in other variables, especially financial variables, reduce the effective quarterly & monthly sample size. Across the sample period, the quarterly (monthly) sample contains an average of 531 (585) stocks with a minimum of 468 (534) and a maximum of 594 (594) stocks.

Table 2 presents Pearson correlation coefficients and shows that ESG scores have the highest correlation (0.42) with Size. Escobar-Saldivar et al. (2025) report a similar correlation between Size and ESG Score of 0.44. This can be attributed to a general strong market cap imbalance between high- and low-ESG firms, which is well documented in the literature and often attributed to the higher amount of financial resources available to larger firms as well as increased stakeholder pressure (see Drempetic et al. 2020, p. 355; Dobrick et al. 2023, p. 1; Gregory 2024, p. 86; Raatz et al. 2024, p. 7). The correlation between ESG scores and ESG Momentum is -0.03.

Table 2: Pearson Correlation Coefficients

This table presents the Pearson correlation coefficients for the period June 2016 to September 2025. *Prof*, *Inv*, *Price Mom*, and *ESG Mom* are abbreviations for *Profitability*, *Investment*, *Price Momentum*, and *ESG Momentum*, respectively. All variables are described in chapter 2.3.

Panel A: Correlation between Cross-Sectional Variables:								
	Beta	Size	Value	Prof	Inv	Price Mom	ESG Score	ESG Mom
Beta		-0.14	0.12	-0.02	0.01	-0.08	-0.02	0.01
Size	-0.14		-0.09	0.00	0.00	0.11	0.42	-0.10
Value	0.12	-0.09		-0.07	-0.02	-0.16	0.10	-0.05
Prof	-0.02	0.00	-0.07		0.00	0.00	-0.03	0.00
Inv	0.01	0.00	-0.02	0.00		0.01	-0.03	0.02
Price Mom	-0.08	0.11	-0.16	0.00	0.01		-0.03	0.03
ESG Score	-0.02	0.42	0.10	-0.03	-0.03	-0.03		-0.03
ESG Mom	0.01	-0.10	-0.05	0.00	0.02	0.03	-0.03	
Panel B: Correlation between Macroeconomic Variables:								
	Dividend Yield		Yield	Term Spread		Default Spread		
Dividend Yield			-0.13	0.19		0.32		
Yield	-0.13			-0.77		-0.03		
Term Spread	0.19		-0.77			-0.07		
Default Spread	0.32		-0.03	-0.07				

Furthermore, there is a strong negative correlation between the term spread and the yield of -0.77. As shown by figure 1, this negative correlation is especially observable starting in 2022. After Russia's invasion of Ukraine in February 2022, the European Central Bank (ECB) raised euro-area interest rates in July 2022, which ended the negative policy rates that have been in place prior (see ECB 2022). As a result, short-term interest rates rose faster than long-term interest rates, resulting in a decline of the term spread. As of September 2025, the term spread has increased again to the same level as before the war broke out, while the 3-month euro area AAA government bond spot rate has stayed at around 2%. There exists support for this observation in the literature. Hall et al. (1992) show that "bonds of different maturity may diverge in the short run" while they "will adjust when the spreads between them deviate from the equilibrium value μ , so that in the long run yields of different maturity will move together". This strong negative correlation might have implications for this analysis, as it is borderline to the maximum limit of the absolute value of 0.8 suggested by Bouraoui & Phisutthiwatcharavong (2015) before multicollinearity might become an issue. However, the corresponding Variance Inflation Factors (VIF) are all below 2.6, which suggests that the resulting inflation of coefficient variances is limited. Following Wooldridge (2013), the practical relevance of multicollinearity depends on the extent to which it increases standard errors, and the present VIF values do not indicate a severe loss of precision.

The brief rise in the dividend yield and default spread during the COVID-19 recession in 2020 is economically intuitive. The dividend-yield increase reflects the fall in equity prices in early 2020, while the trailing twelve-month dividends adjusted only gradually. The widening default spread reflects heightened perceived default risk for lower-rate corporate bonds during the pandemic shock (see Apergis et al. 2022, p. 1).

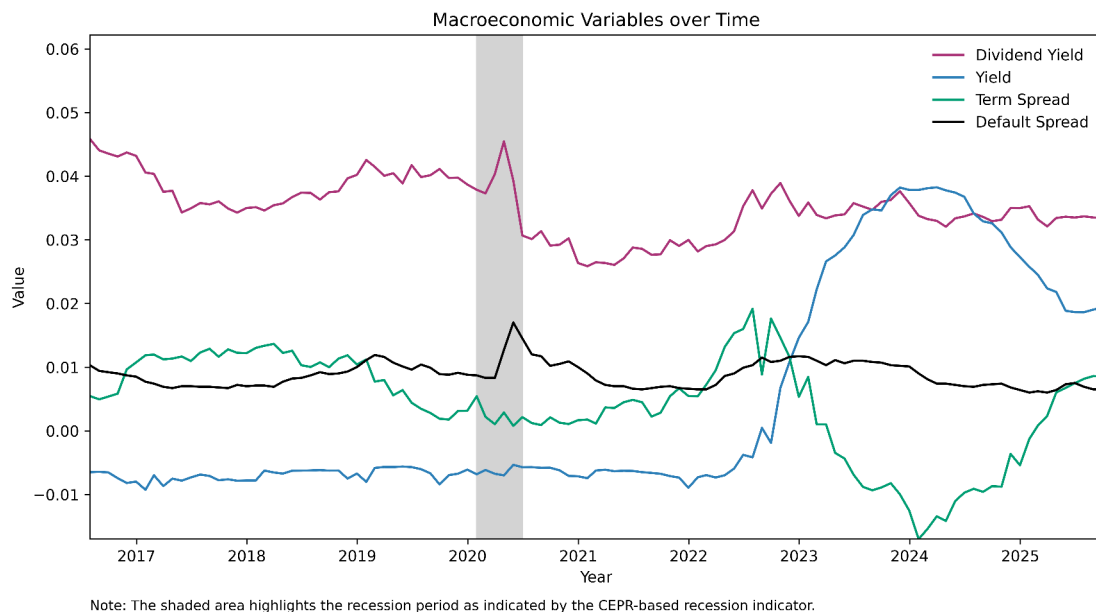


Figure 1: Macroeconomic Variables over Time

4 Empirical Results & Discussion

4.1 Relation between ESG momentum and Stock Returns among STOXX Europe 600 stocks

Table 3 presents the main results of the pooled OLS, fixed effects, random effects, and two-step difference generalized method of moments analysis between June 2016 and September 2025, reported in columns *POLS*, *FE*, *RE*, and *GMM (I)*, respectively. For pooled OLS, fixed effects and random effects, the coefficients are similar. In fact, pooled OLS and random effects produce the exact same results. Mechanically, this occurs because the raw estimate of the variance of the individual effect $\hat{\sigma}_a^2$ is negative. The between-regression residual variance ($\frac{SSR}{N-k}$) is smaller than the time-averaged idiosyncratic variance $\frac{\hat{\sigma}_u^2}{T}$. The estimator is therefore truncated at zero (as explained in section 2.3.1.1) resulting in $\hat{\theta}$ also being zero and the random effects transformation coinciding with pooled OLS. To assess whether this result is due to a specific component of the model specification, I re-estimate the model repeatedly while dropping one regressor at the time. Only dropping price momentum yielded a positive estimate of the random-effect variance. This suggests that, in this sample, price momentum absorbs a substantial part of the between-stock variation in returns, meaning that the stock-level average price momentum is associated with the stock-level average return across firms. This interpretation is consistent with Conrad & Kaul (1998) who argue that price momentum profits can arise from the cross-sectional variation in stock mean returns. They say that a price momentum strategy is, on average, equal to buying high-mean stocks and selling low-mean stocks and that as long as there is some cross-sectional separation in the mean returns of stocks, a price momentum strategy will be profitable. Consequently, if price momentum is informative about which stocks are high-mean versus low-mean stocks, then the average price momentum would line up with the average stock return and therefore reduce the residual variance in the between regression.

The Hausman (1978) test rejects its null hypothesis at the 1% level, indicating that fixed effects is preferred to random effects. In the fixed effects model, the ESG score coefficient is negative (-0.0005) and statistically significant at the 1% level. ESG momentum is positive (0.0002) and significant at the 10% level. This suggests that larger ESG scores are associated with lower quarterly returns while increases in ESG scores over the previous 12 months are associated with higher quarterly returns. Specifically, a 10-point higher ESG score is associated with a 0.5 percentage points lower quarterly return and a 10-point increase in ESG score over the previous 12 months is associated with about 0.2 percentage points higher quarterly return. This finding is consistent with Escobar-

Saldivar et al. (2025), who also report a statistically significant negative coefficient for ESG scores and a significantly positive coefficient for ESG momentum. However, while they find ESG momentum to be significant at the 1% level, it is only significant at the 10% level in my analysis which indicates weaker evidence in my sample.

Table 3: Results of Main Specifications POLS, FE, RE, GMM

This table presents the estimated coefficients for the period from June 2016 to September 2025 of the multi-factor models in Equations 1 and 8 obtained by performing pooled OLS (POLS), fixed effects (FE), random effects (RE), and two-step difference generalized method of moments (GMM) as described in chapters 2.3.1.1 and 2.3.1.2. The constant, industry, country, and time dummy coefficients are omitted for better readability but can be viewed in appendix A.1. Independent variables are winsorized at the 0.5% and 99.5% level. Entity clustered t-statistics are shown in parenthesis for POLS, FE, and RE. Windmeijer-corrected z-statistics are shown in parenthesis for GMM. *, **, *** show statistical significance at the 10%, 5%, and 1% level.

	POLS	FE	RE	GMM (1)
Observations	20,184	20,184	20,184	17,203
R-Squared (Within)	0.6510	0.3246	0.6510	
R-Squared (Between)	0.4323	0.2007	0.4323	
R-Squared (Overall)	0.4418	0.1657	0.4418	
R_{q-1}				0.0167 (0.48)
Beta	0.0031*** (2.85)	0.0034*** (2.62)	0.0031*** (2.85)	0.0054 (0.94)
Size	0.0075*** (11.97)	0.0358*** (8.19)	0.0075*** (11.97)	0.0306 (0.76)
Value	-0.0155*** (-6.64)	-0.0421*** (-5.69)	-0.0155*** (-6.64)	-0.0813** (-2.08)
Prof	-0.0094*** (-3.41)	-0.0106** (-2.15)	-0.0094*** (-3.41)	0.057 (1.52)
Inv	0.0101 (1.12)	0.0082 (0.93)	0.0101 (1.12)	-0.5012 (-1.08)
Price Mom	0.3343*** (61.49)	0.3046*** (46.50)	0.3343*** (61.49)	-0.0453 (-1.07)
ESG	-0.0001 (-1.49)	-0.0005*** (-3.78)	-0.0001 (-1.49)	-0.0019** (-2.33)
ESG Mom	0.0001 (0.53)	0.0002* (1.68)	0.0001 (0.53)	0.0012* (1.85)

Notes on GMM:

Hansen p-value: 0.231; AR(1) p-value: 0.000; AR(2) p-value: 0.183

Instruments are collapsed. Instrument Count: 52

Lags used: $r_{t-1} = L.2-L.3$; Price Momentum = L.4-L.5; All others = L.3

This paper extends the approach by Escobar-Saldivar et al. (2025) by additionally estimating a two-step difference generalized method of moments (GMM) model. The GMM coefficients of ESG score and ESG momentum exhibit the same sign pattern as those obtained from the fixed effects model, but they are larger in magnitude. The coefficient on ESG score is negative (-0.0019) and significant at the 5% level. The coefficient on ESG momentum is positive (0.0012) and significant at the 10% level. Based

on the GMM results, a 10-point higher ESG score is associated with a 1.9 percentage points lower quarterly return and a 10-point increase in ESG score over the previous 12 months is associated with about 1.2 percentage points higher quarterly return. The pattern of larger absolute values for the two ESG coefficients could be interpreted as the fixed effects estimates being biased towards zero due to endogeneity. If companies with strong prior financial performance have more internal resources to build and sustain higher ESG activity, and therefore higher ESG score levels, this would be consistent with Waddock & Graves (1997), who argue that firms with strong financial performance have more resources available to invest in CSR activities. In that case, ESG scores may partly reflect prior firm performance. As a result, this positive correlation might offset the true negative association between ESG levels and returns and push the fixed effects estimate upward towards zero. Similarly, if, as documented by Shackleton et al. (2022), firms respond to poor stock performance by increasing their environmental and social activity, then positive ESG momentum might be associated with poor performance pulling the FE coefficient downward, making the positive ESG momentum effect look smaller than it really is.

Among the control variables, profitability is negative and significant in POLS/RE and FE, whereas investment is positive but insignificant in all three models. Price momentum has the largest coefficient and is strongly significant at the 1% level. In GMM, however, profitability turns positive and becomes insignificant, investment turns negative and increases sharply in absolute value while remaining insignificant, and price momentum weakens substantially, turning negative and insignificant. Beta and size also lose statistical significance in GMM. Value remains negative and statistically significant at the 5% level in GMM. Overall, these shifts suggest that the beta, size, profitability and especially price momentum effects in the static panel models are not fully robust once allowing for dynamic endogeneity. This change in the control-variable estimates makes it particularly important to assess whether the GMM specification is econometrically credible, which I evaluate using the Hansen test and Arellano-Bond serial correlation tests. The Hansen test p-value of 0.231 does not reject instrument validity. However, it is fairly close to the upper limit of 0.25 suggested by Roodman (2009) where one should be alert of potentially weak power of the Hansen test. After collapsing the instruments, the instrument count is 52 which is below the number of individual stocks in the sample and therefore does not immediately suggest severe instrument proliferation (see Roodman 2009, p. 99). However, it is large enough that some risk of overfitting endogenous variables remains. As explained in section 2.3.1.2, AR(1) is expected to be statistically significant, so one needs to look at the AR(2) p-value. The AR(2) p-value of 0.183 indicates that there is no evidence of second-order serial correlation in the differenced residuals. Therefore, the validity of the lags cannot be rejected. As conservative benchmark, no variable is treated as strictly exogenous and instead instrumented using

internal lags¹. The lagged quarterly return is instrumented with lags 2 and 3. Beta, size, value, profitability, investment, ESG, and ESG momentum are instrumented using lag 3 and price momentum uses lags 4 and 5. To assess robustness, I re-estimate GMM under six alternative specifications (GMM (2) - GMM (7)) presented in table 4 with specification changes highlighted in bold.

GMM (2) covers the entire period from June 2016 to September 2025 but uses deeper lags. Beta, size, value, profitability, investment, ESG and ESG momentum are instrumented using lag 4. Price momentum is instrumented using lag 6. GMM (3) instruments price momentum with lag 3 instead of lags 4 and 5. GMM (4) also instruments price momentum with lag 3 and treats profitability as an not-internally instrumented IV-style variable. Profitability uses an annual operating income measure which makes it less likely to react to a single quarter's idiosyncratic return shock (cf. Fama & French 2006, p. 495) and could therefore be a plausible candidate for a less conservative treatment. GMM (5) through GMM (7) use the same lags as GMM (1) but are estimated over different sub periods. GMM (5) looks at the period between June 2016 and June 2019. GMM (6) spans September 2019 to September 2022, covering the COVID-19 period. GMM (7) estimates the period between December 2022 and September 2025.

The results of GMM (2) through GMM (4) show that ESG momentum remains positive and ESG negative. However, ESG momentum only remains significant at the 10% level for GMM (3) and GMM (4). Similarly, the ESG score coefficient is significant at the 5% level only for GMM (3) and GMM (4). GMM (2) deepens the internal lags used as internal instruments which may reduce concerns that the instruments remain too close to the current error term. However, the larger Hansen p-value of 0.734 possibly indicates a loss of test power caused by weaker internal instruments. Wintoki et al. (2012) explain that deeper lags are less informative about the endogenous regressors. GMM (3), on the other hand, uses shallower lags as internal instruments for price momentum and raises concerns of second-order serial correlation in the differenced residuals with an AR(2) p-value of 0.085, which could render some lags invalid as instruments (see Roodman 2009, p. 119). The shallower lags for price momentum may improve instrument relevance, but in this specification it also appears to leave the instrument set more exposed to residual serial correlation. The results of GMM (2) and GMM (3) highlight the trade-off between instrument exogeneity and relevance. Looking at GMM (4), treating profitability as an IV-style variable improves the AR(2) p-value slightly compared to GMM (3). The coefficients of ESG and ESG momentum are -0.0020 and 0.0013, respectively and therefore almost identical to GMM (1) and GMM (3).

¹To conserve space and since the main variables of interest are the coefficients of ESG score levels and ESG Momentum, endogeneity concerns of the control variables are not discussed here. See e.g. Wintoki et al. (2012), Chinloy & Imes (2025), Schultz et al. (2010) for further discussions.

Table 4: Results of Robustness Specifications of GMM

This table presents the estimated coefficients for six different specifications of the two-step difference generalized method of moments model in Equation 8. Any specification changes made to the base model GMM (1) presented in table 3 are highlighted in bold. The time dummy coefficients are omitted for better readability but can be viewed in appendix A.2. Independent variables are winsorized at the 0.5% and 99.5% level. Windmeijer-corrected z-statistics are shown in parenthesis. *, **, *** show statistical significance at the 10%, 5%, and 1% level.

	GMM (2)	GMM (3)	GMM (4)	GMM (5)	GMM (6)	GMM (7)
Sample Period	06/2016 - 09/2025	06/2016 - 09/2025	06/2016 - 09/2025	06/2016 - 06/2019	09/2019 - 09/2022	12/2022 - 09/2025
Lags:						
R_{q-1}	2-3	2-3	2-3	2-3	2-3	2-3
Beta	4-4	3-3	3-3	3-3	3-3	3-3
Size	4-4	3-3	3-3	3-3	3-3	3-3
Value	4-4	3-3	3-3	3-3	3-3	3-3
Prof	4-4	3-3	N/A	3-3	3-3	3-3
Inv	4-4	3-3	3-3	3-3	3-3	3-3
Price Mom	6-6	3-3	3-3	4-5	4-5	4-5
ESG	4-4	3-3	3-3	3-3	3-3	3-3
ESG Mom	4-4	3-3	3-3	3-3	3-3	3-3
R_{q-1}	0.0114 (0.35)	0.0165 (0.49)	0.0216 (0.63)	0.0147 (0.13)	-0.0843 (-0.80)	0.959 (0.60)
Beta	0.027*** (3.18)	0.0085 (1.46)	0.0066 (1.11)	0.0052 (0.24)	0.0342 (0.83)	0.0050 (0.38)
Size	0.008 (0.19)	0.0402 (1.04)	0.0510 (1.26)	0.0754 (0.39)	0.0252 (0.06)	-0.0119 (-0.07)
Value	-0.0809** (-2.19)	-0.0499 (-1.32)	-0.0569 (-1.43)	-0.1913 (-0.67)	-0.6564* (-1.75)	-0.2211 (-1.56)
Prof	0.0226 (0.57)	0.0400 (1.30)	-0.0119 (-0.96)	-0.0521 (-0.23)	0.3996** (2.01)	0.1823 (1.39)
Inv	0.0285 (0.11)	-0.5163 (-1.10)	-0.5936 (-1.22)	0.5766 (1.00)	-2.3360** (-2.23)	0.7481 (0.36)
Price Mom	-0.0317 (0.85)	-0.0373 (-0.88)	-0.0486 (-1.12)	-0.0396 (-0.29)	0.0261 (0.17)	-0.2145 (-1.03)
ESG	-0.0010 (-1.56)	-0.0019** (-2.38)	-0.0020** (-2.53)	0.0001 (0.02)	-0.0153** (-2.01)	-0.0106 (-1.51)
ESG Mom	0.0003 (0.28)	0.0012* (1.75)	0.0013* (1.89)	0.0020 (1.58)	0.0021 (0.59)	0.0002 (0.007)
Instrument Count	42	45	45	20	20	19
P-values:						
Hansen	0.734	0.133	0.172	0.148	0.139	0.907
AR(1)	0.000	0.000	0.000	0.003	0.009	0.004
AR(2)	0.238	0.085*	0.118	0.996	0.163	0.032**

GMM (5) through GMM (7) show that the results from the benchmark specification GMM (1) are not robust to different time periods while retaining the same lag structure. In the pre COVID-19 sample GMM (5), June 2016 to June 2019, the post estimation diagnostics are satisfactory, yet neither ESG score nor ESG momentum are statisti-

cally significant. In the period sample that includes the COVID-19 pandemic GMM (6), September 2019 to September 2022, the diagnostics also remain acceptable and the ESG score coefficient becomes strongly negative (-0.0153) and statistically significant at the 5% level. ESG momentum however, remains insignificant. This suggests that during this time period the cross-section of returns was more strongly related to ESG levels than to recent ESG upgrades. By contrast, the last specification period GMM (7), December 2022 to September 2025, is not econometrically convincing. The Hansen p-value rises to 0.907 which may indicate weak test power, and the AR(2) p-value of 0.032 implies that the validity of the chosen lags can no longer be maintained. Accordingly, GMM (7) should be treated as a failed robustness check rather than as valid evidence.

Overall, the fixed-effects and two-step difference GMM results display the same sign pattern documented by Escobar-Saldivar et al. (2025) for U.S. stocks. Therefore, research question **Q1** can be answered with a qualified yes in terms of sign consistency, but not in terms of uniformly strong statistical robustness: ESG levels seem to be negatively related to stock returns, whereas ESG momentum appears to be positively related to stock returns. Together, the results are consistent with the interpretation that the two variables proxy for different economic mechanisms. ESG levels may reflect lower risk and therefore lower required returns consistent with prior research (see e.g., El Ghoul et al. 2011, p. 2388; Ciciretti et al. 2023, pp. 104, 106; Escobar-Saldivar et al. 2025, p. 9). ESG momentum could be interpreted to capture repricing during the period in which investors incorporate new ESG information into valuations (see Koziol & Roßmann 2022, pp. 136–137; Nagy et al. 2016, p. 113; Galema & Gerritsen 2025, p. 1). However, the evidence presented here for the European sample is weak since the fixed-effects and GMM (1) coefficients on ESG momentum are only statistically significant at the 10% level. Additionally, the results are sensitive to the choice of instrument lags and sample period. Furthermore, it is worth mentioning that there exists evidence that Refinitiv ESG scores are subject to historical rewriting. Berg et al. (2020) argue that the data is rewritten by the rating vendor to retroactively strengthen the link between ESG scores and returns as they find a positive link between ESG scores and stock returns in the rewritten data but fail to observe such a relationship in the initial data. If true, the analysis could be, at least partially, based on information that was not available to investors at the time and could bias the results. This raises concerns about the general validity of Refinitiv ESG ratings for empirical research and should warrant caution when interpreting the results of any study using Refinitiv ESG scores. Future research should therefore perform the here presented analysis using ESG ratings from different providers that don't exhibit historical rewriting to validate the results. Nevertheless, together with the gaps identified in the literature review, the differing results across GMM (5) through GMM (7) highlight the need for further research on the relation between ESG momentum and the business cycle, which is examined in the next section.

4.2 Business Cycle Analysis

This chapter examines ESG momentum payoffs over time and across business-cycle phases, and tests if macro-predicted and explained return components account for them.

4.2.1 ESG Momentum Payoffs Over Time

Figure 2 shows the cumulative value-weighted and equal-weighted ESG momentum payoffs, along the cumulative returns of other factors, including size, value, profitability, investment, and price momentum, from the Fama-French data library over time. The shaded area highlights the CEPR-based recession period.

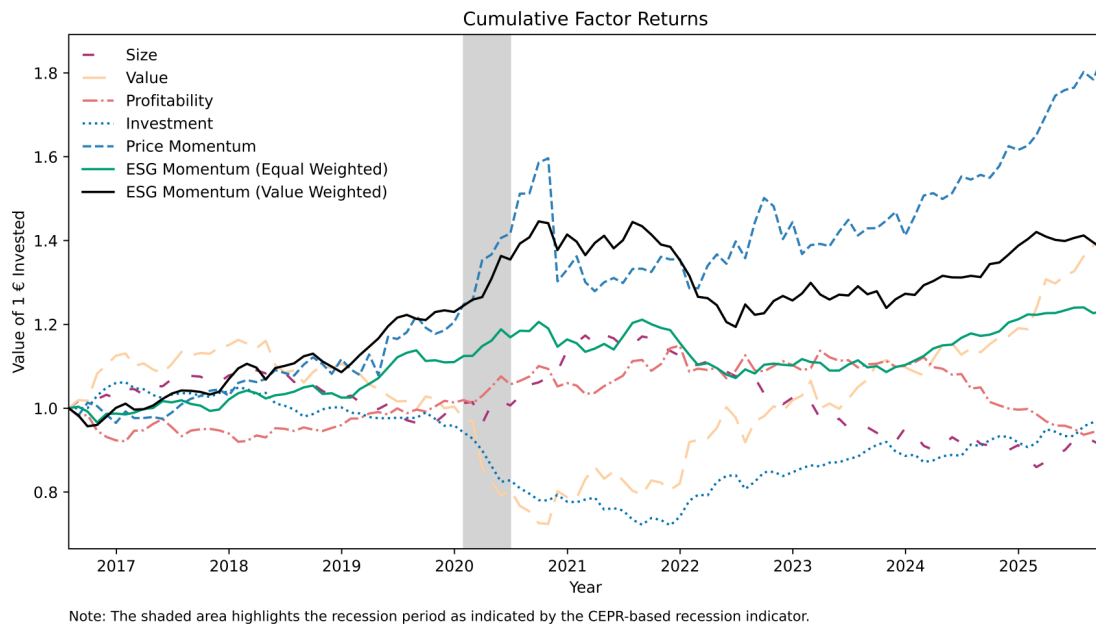


Figure 2: Cumulative Factor Returns

It can be seen that both the equal-weighted and the value-weighted strategies have yielded positive cumulative returns between June 2016 and September 2025. The value-weighted ESG momentum strategy delivered the second-highest cumulative return among the plotted strategies, trailing only price momentum. However, this performance is not consistent over time. The gains mainly accrue during the period from June 2016 to September 2020 and decrease or remain flat afterwards. The equal-weighted ESG momentum strategy has underperformed the value-weighted ESG momentum strategy suggesting that the ESG momentum payoffs are higher among larger stocks. A possible explanation is that ESG momentum is more strongly priced among large-cap stocks because institutional investors, as shown by Ferreira & Matos (2008), tend to prefer large firms. This interpretation is consistent with evidence that institutional ownership is positively associated with environmental and social performance (see Dyck et al. 2019, p. 693) and that institutional investors can induce companies to increase their corporate

social responsibility ratings (see Chen et al. 2020, p. 492). It is however also plausible that small stocks have generally realized lower returns and the equal-weighted strategy captures this underperformance. Figure 2 supports this by showing that the overall cumulative size factor performance has been negative over the entire sample period.

Table 5: ESG Momentum Payoffs Classified by Business Cycle

This table presents the time-series average equal-weighted and value-weighted payoffs from a strategy that buys (sells) the 20% of stocks with the highest (lowest) ESG momentum (detailed explanations are provided in section 2.3.2.1). The classification into recession and non-recession periods is based on the CEPR-EABCN Euro Area Business Cycle dates (<https://eabcn.org/dbc/graphs>). t-statistics (in parenthesis) are based on Newey & West (1987) heteroskedasticity and autocorrelation consistent (HAC) standard errors. The lag length is chosen as $g = [4(n/100)^{2/9}]$ where n is the number of monthly observations in the relevant time-series sample. *, **, *** show statistical significance at the 10%, 5%, and 1% level.

Period	Months	Recession	Equal Weighted ESG Momentum Payoff	Value Weighted ESG Momentum Payoff
07/2016 - 12/2019	42	No	0.002366 (1.12)	0.004682* (1.87)
01/2020 - 06/2020	6	Yes	0.008770** (2.18)	0.016401*** (3.41)
07/2020 - 09/2025	63	No	0.000900 (0.48)	0.00047 (0.20)

Table 5 presents the time-series average equal-weighted and value-weighted payoffs to an ESG momentum strategy during three different business cycle periods. The periods from July 2016 to December 2019 and from July 2020 to September 2025 are non-recession periods, whereas the period from January 2020 to June 2020 was a recession period as indicated by the CEPR-EABCN Euro Area business cycle dates. It can be seen that all average payoffs were positive during each period although the average payoffs between July 2020 and September 2025 were much smaller than during the other periods. For the equal-weighted strategy, only the payoffs during the recession period were statistically significant at the 5% level. The value-weighted payoffs were both statistically significant before and during the recession but no longer significant after the recession period. Both the equal-weighted and value-weighted payoffs were highest during the recession window. This is broadly consistent with Yoo et al. (2021), who report that during the COVID-19 pandemic, increases in ESG scores, especially the environmental score, were associated with higher returns. That said, their evidence regards firm-level ESG score changes during the pandemic rather than the profitability of an ESG momentum long-short strategy. Furthermore, this interpretation should be treated cautiously, since the sample contains only two non-recession and only one recession period which makes the results difficult to generalize. The recession period also only lasted for six months which makes the interpretation of the results particularly fragile. Overall, the results of this section are suggestive rather than conclusive.

4.2.2 Predicted Returns Across ESG Momentum Portfolios

Table 6 presents the pooled by macro variables predicted median stock returns of the portfolio with the 20% lowest ESG momentum stocks P1 and the portfolio with the 20% highest ESG momentum stocks P5 for the event months 0 and 1.

Table 6: Median Predicted Stock Returns Around The Formation Period

This table presents the pooled median predicted stock returns for the portfolios P1 and P5 for the formation month (0) and the holding month (1). The methodology for obtaining these results is described in chapter 2.3.2.3. The results are reported separately for training windows of 36, 48, and 60 months. It further shows the percentage of months where the difference between the median predicted returns of P5 and P1 was positive ($\% > 0$) as well as the p-values of a nonparametric two-sided and a nonparametric one-sided greater sign test. *, **, *** indicate statistical significance at the 10%, 5%, and 1% level.

	36 Months		48 Months		60 Months	
	0	1	0	1	0	1
P1	0.0115	0.0117	0.0019	0.0007	0.0007	-0.00001
P5	0.0138	0.0144	0.0038	0.0029	0.0032	0.0030
% > 0	62.16%	59.46%	64.52%	58.06%	76.00%	80.00%
P-Value Two-Sided	0.047**	0.130	0.030**	0.253	0.000***	0.000***
P-Value Greater	0.024**	0.065*	0.015**	0.126	0.000***	0.000***

Although the absolute level of the pooled median predicted returns varies materially across the training windows of 36, 48, and 60 months, the median predicted returns of portfolio P5 are higher than those of portfolio P1 in both event month 0 and event month 1. This indicates that stocks with high ESG momentum tend to have higher by macroeconomic variables predicted returns than stocks with low ESG momentum, both in the formation month and in the subsequent holding month. As this pattern is observable in both event months, the by macro variables predicted return component is not limited to the sorting month, but extends into the month in which the ESG momentum strategy is actually held. However, the sign-test results show that this ordering is not equally strong across all training windows. With a 36-month training window, the share of months in which the cross-sectional median predicted return of P5 exceeds that of P1 is 62.16% in event month 0 and 59.46% in event month 1. The two-sided sign test is statistically significant at the 5% level only for event month 0. The one-sided greater sign test is statistically significant at the 5% level for event month 0 and at the 10% level for event month 1. This suggests that the null hypothesis that the cross-sectional median predicted return of P5 exceeds that of P1 in at most half of the months can be rejected for both event months, although the evidence is weaker for event month 1. Similarly, for the 48-month training window, the two-sided sign test is again only statistically

significant for event month 0. The one-sided greater sign test remains significant for event month 0 but is no longer significant for event month 1. For the 60-month training window, the results become stronger in both event months. The percentage of positive P5 minus P1 median differences increases to 76% in event month 0 and 80% in event month 1. Both sign tests are statistically significant at the 1% level in both event months.

The stronger evidence for the 60-month training window may reflect a trade-off in the rolling regression design. On the one hand, a longer training window uses more historical observations to estimate the stock-level regression, which generally reduces the variance of the coefficient estimates (see Wooldridge 2013, p. 54) and can therefore produce less noisy predicted returns. On the other hand, requiring 60 months of estimation data reduces the usable sample as predicted returns can only be generated for stocks and months with a sufficiently long return history. Given the relatively short overall sample period from June 2016 to September 2025, this comes at the cost of discarding early observations. In addition, the stronger evidence for the 60-month window may reflect not only greater estimation precision, but also a change in sample composition as the exclusion of earlier observations changes the set of months and stocks entering the comparison. By contrast, shorter training windows keep more observations in the sample, but the estimated coefficients may be less precise. As a result, the predicted-return differences between ESG momentum portfolios may appear less consistent.

Overall, the results carefully suggest that the component of returns predicted by macroeconomic variables tends to be higher for high ESG momentum stocks than for low ESG momentum stocks. This pattern is already visible at portfolio formation and persists into the subsequent holding month. Consistent with the intuition of Chordia & Shrivastava (2002), this indicates that differences in by macroeconomic predicted returns across ESG momentum portfolios may account for part of the profitability of the ESG momentum strategy. Hence, the results are in line with the interpretation that ESG momentum portfolios may partly sort stocks on differences in time-varying expected returns associated with the macroeconomic conditions, rather than solely on ESG-specific information unrelated to the business cycle. At the same time, because significance is not uniform across all training windows, this evidence should be interpreted cautiously and at most as supportive rather than perfectly robust. Furthermore, the fitted values of the stock-level regression includes the intercept term and may therefore reflect not only the return component associated with the macroeconomic variables, but also persistent stock-specific differences in returns. The next section looks at the macro-variables predicted component by adjusting realized returns for the predicted component, excluding the intercept, and then tests if a residual ESG momentum payoff remains.

4.2.3 ESG Momentum Payoffs Using Adjusted Returns

This analysis asks whether ESG momentum generates different payoffs after removing the component of returns that is predictable from macroeconomic conditions. If the adjusted payoff is different, this would suggest that the ESG momentum strategy partly captures business cycle related expected returns rather than only ESG-specific repricing. Table 7 presents the equal-weighted and value-weighted ESG momentum payoffs after adjusting for returns predicted by the macro variables dividend yield, yield, term spread, and default spread. Panel A defines predicted returns using only the fitted macro-variable component, so adjusted returns include the intercept and residual. Panel B also includes the intercept in predicted returns, so adjusted returns capture only the residual component. Panel C reports the corresponding raw ESG momentum payoffs for the exact subsample of stock-months used in Panels A and B. It serves as the appropriate benchmark for assessing the effect of the macroeconomic adjustment. The average raw payoffs are positive across all three training windows consistent in sign with the results obtained in section 4.2.1. However, none are statistically significant at any conventional significance level. This follows the earlier results that significance levels are dependent on the analyzed time period and considered sample. By construction, this analysis excludes for each training window of 36, 48, or 60 months, the first W holding months as well as any stock for which there are less than 24 monthly observations within the corresponding model training window. Consequently, the analysis of differences between Panel C and Panels A or B is based on an already weak raw payoff on the restricted sample.

For the equal-weighted strategy in Panel A, adjusted payoffs are not robust across training windows: they are positive and marginally significant for 36 months, positive but insignificant for 48 months, and negative and insignificant for 60 months. The sign tests are also insignificant. For the 36-month window the raw equal-weighted payoffs are smaller than the adjusted payoffs, suggesting that the macro-predicted component is negative on average. By contrast, the value-weighted adjusted payoffs are negative and statistically significant across all three training windows. In addition, the percentages of months with positive value-weighted P5–P1 adjusted payoffs are low, ranging from 21.57% to 32.00%. This implies that the intercept plus residual component is persistently negative for the value-weighted ESG momentum strategy. Since Panel C shows that the raw value-weighted payoffs on the same subsample are positive, although insignificant, the negative adjusted payoffs in Panel A suggest that the macro-predicted component is positive and offsets an otherwise negative non-macro component. Specifically, the results of Panel B imply that the negative component in Panel A appears to be driven primarily by the intercept rather than by the residual component. In Panel B, where the intercept is included in the predicted return and the adjusted return therefore cap-

tures only the residual component, the adjusted payoffs are less negative than in Panel A and statistically insignificant. The percentage of months with positive adjusted ESG momentum payoffs is also larger and closer to 50% similar to Panel C. The stronger negative results for value-weighted portfolios in Panel A indicate that the non-macro component may be unfavorable for large-cap stocks once the positive macro-predicted component is removed. However, because the raw payoffs in the subsample are already small and insignificant, the findings should be interpreted carefully. At most, the evidence suggests that time-varying expected returns may contribute to ESG momentum payoffs, but it does not establish a robust macroeconomic explanation.

Table 7: ESG Momentum Payoffs Adjusted for Macroeconomic Variables

This table reports equal- & value-weighted ESG momentum payoffs using adjusted returns, defined as the realized return minus the predicted return. Thus, the “P5 - P1” columns show the part of payoffs unexplained by macroeconomic variables. The methodology is described in Chapter 2.3.2.4. The results are reported separately for training windows of 36, 48, and 60 months. Panel A (B) shows the adjusted payoffs with the intercept included (excluded) in (from) the adjusted stock return. Panel C presents the raw payoffs from the ESG momentum strategy for the subsample of stock-months used in Panel A and B. t-statistics (in parenthesis) are based on Newey & West (1987) heteroskedasticity and autocorrelation consistent (HAC) standard errors. The lag length is chosen as $g = [4(n/100)^{2/9}]$ where n is the number of monthly observations in the relevant time-series sample. The column titled “% > 0” gives the percentage of P5 - P1 that are positive, and p-values from the two-sided sign test are given in parenthesis below the percentage value. *, **, *** indicate statistical significance at the 10%, 5%, and 1% level.

	Equal Weighted Payoffs		Value Weighted Payoffs	
	P5 - P1	% > 0	P5 - P1	% > 0
Panel A: Adjusted Payoffs Includes Intercept				
36 Months	0.0122* (1.75)	56.58% (0.3557)	-0.0166** (-2.54)	32.00%*** (0.0024)
48 Months	0.0035 (0.65)	52.38% (0.8013)	-0.0169** (-2.57)	28.57%*** (0.0009)
60 Months	-0.0023 (-0.49)	49.02% (1.0000)	-0.0190*** (-3.08)	21.57%*** (0.0001)
Panel B: Adjusted Payoffs Excludes Intercept				
36 Months	-0.0006 (-0.35)	48.00% (0.8176)	-0.0012 (-0.59)	49.33% (1.000)
48 Months	0.0007 (0.40)	57.14% (0.3135)	0.0011 (0.4867)	58.73% (0.2074)
60 Months	0.0001 (0.06)	50.98% (1.0000)	-0.0005 (-0.17)	50.98% (1.0000)
Panel C: Raw Payoffs For The Subsample Of Stock-Months Used In Panel A And B				
36 Months	0.0010 (0.64)	58.67% (0.1654)	0.0014 (0.71)	54.67% (0.4887)
48 Months	0.0014 (0.85)	58.73% (0.2074)	0.0008 (0.34)	53.97% (0.6147)
60 Months	0.0019 (1.05)	58.82% (0.2624)	0.0005 (0.19)	50.98% (1.0000)

4.2.4 ESG Momentum Payoffs Regressed on Macroeconomic Variables

Table 8 presents the time-series average coefficients from rolling regressions of the equal-weighted and value-weighted ESG momentum payoffs on the four macroeconomic variables lagged by one month. Panel A (B) shows the results for the equal-weighted (value-weighted) payoffs. The average payoff unexplained by the macro variables *RES* is computed as the intercept *INT* plus the residual. The results show that *RES* is negative in all specifications and statistically significant for the 60-month equal-weighted specification and for all value-weighted specifications. Moreover, the fraction of months with positive unexplained payoffs declines strongly with the training window and falls to only 3.92% in the 60-month value-weighted specification. Together this suggests that the positive component of ESG momentum payoffs appears to be largely associated with the macroeconomic variables and once the predicted component is removed, the average unexplained components are negative. Since *RES* is very close to *INT*, the unexplained payoff seems to be driven mainly by the intercept rather than by a negative residual. Overall, the results do not support the existence of a positive ESG momentum premium after conditioning on macroeconomic variables. Rather, they suggest that the positive payoff component is at least partly associated with macroeconomic conditions captured by the macro variables.

The statistically significant positive dividend yield coefficients in Panel A and Panel B suggest that ESG momentum payoffs tend to be larger when the market dividend yield is high. As already briefly mentioned in chapter 3, a higher dividend yield may proxy for higher discount rates and consequently lower stock valuations as the trailing twelve-month dividend payments of the STOXX Europe 600 index adjust only gradually while equity prices may adjust more quickly. The results imply that if ESG improvements are interpreted by investors as signals of lower non-financial risk (see Escobar-Saldivar et al. 2025, p. 16), then those signals may matter more when aggregate discount rates, proxied by higher dividend yields, are high. This interpretation is directionally consistent with Yoo et al. (2021), who show that increases in ESG scores were associated with higher returns during the COVID-19 pandemic. It is also broadly consistent with the spike in dividend yield shown in figure 1 during the COVID-19 pandemic and the positive ESG momentum payoffs reported in table 5. The 3-month euro area AAA government bond spot rate enters the regression both directly via the yield variable and indirectly through the term spread, since the term spread is defined as the difference between the annualized 10-year and 3-month spot rates. Thus, holding the term spread constant, the results imply that a higher yield is equivalent to a higher level of interest rates. Therefore, the positive yield coefficients suggest that ESG momentum payoffs tend to be higher when the level of rates is higher. The negative term spread coefficients indicate that ESG mo-

momentum payoffs are lower when the yield curve is steeper and stronger when the yield curve is inverted. Together, one possible interpretation is that ESG momentum payoffs may be higher under macroeconomic conditions in which the short end of the yield curve is high relative to the long end and in which short-term rate levels are generally elevated. This is coherent with the idea that such monetary conditions expose differences in financing vulnerability as firms with high debt rollover needs may feel the effect of higher interest rates more strongly (see Abbas et al. 2025, p. 23). It is also coherent with evidence reported by Ge & Liu (2015) that firms with better CSR performance obtain better credit ratings, lower bond yield spreads, and fewer covenant restrictions. Taken together, these findings suggest the possibility that ESG score changes are correlated with characteristics that become more valuable when monetary conditions tighten, such as perceived non-financial risk or lower financing vulnerability. Complementary evidence is provided by Lins et al. (2017) who find that companies with high CSR activity raised more debt during the 2008-2009 financial crisis. Further research could investigate if the apparent positive association of ESG momentum payoffs with the yield and the negative association with the term spread are driven primarily by P1 underperformance or P5 outperformance.

Table 8: ESG Momentum Payoffs Regressed on Macroeconomic Predictor Variables

This table presents the time-series average coefficients from regressions of the equal-weighted (Panel A) and value-weighted (Panel B) ESG momentum payoffs on the lagged macroeconomic variables dividend yield (*DIV*), yield (*YLD*), term spread (*TERM*), and default spread (*DEF*). *RES* is the average unexplained payoff computed as intercept (*INT*) plus the residual. The methodology for obtaining these results is described in chapter 2.3.2.5. The results are reported separately for training windows of 36, 48, and 60 months. t-statistics (in parenthesis) are based on Newey & West (1987) heteroskedasticity and autocorrelation consistent (HAC) standard errors. The lag length is chosen as $g = [4(n/100)^{2/9}]$ where n is the number of monthly observations in the relevant time-series sample. The column titled "% *RES* > 0" gives the percentage of *RES* that are positive, and p-values from the two-sided sign test are given in parenthesis below the percentage value. *, **, *** indicate statistical significance at the 10%, 5%, and 1% level.

	RES	% RES > 0	INT	DIV	YLD	TERM	DEF
Panel A: Equally Weighted ESG Momentum Payoffs							
36	-0.003	52.0%	-0.0004	1.03***	3.66***	-0.09	-0.77*
months	(-0.29)	(0.818)	(-0.04)	(3.47)	(3.42)	(-0.98)	(-1.92)
48	-0.011	38.1%*	-0.008	1.06***	2.84***	-0.11*	-0.73***
months	(-1.26)	(0.077)	(-0.96)	(7.27)	(3.12)	(-1.76)	(-3.28)
60	-0.02***	27.5%***	-0.01*	0.89***	1.70**	-0.11***	-0.36**
months	(-2.71)	(0.002)	(-1.92)	(11.08)	(2.30)	(-2.85)	(-2.03)
Panel B: Value Weighted ESG Momentum Payoffs							
36	-0.04***	30.7%***	-0.04***	1.36***	1.76*	-0.25***	0.84
months	(-2.89)	(0.001)	(-2.60)	(2.81)	(1.92)	(-2.72)	(1.41)
48	-0.04***	12.7%***	-0.04***	1.31***	1.42**	-0.21**	1.11***
months	(-4.81)	(0.000)	(-4.35)	(4.64)	(2.00)	(-2.44)	(2.67)
60	-0.04***	3.92%***	-0.04***	1.09***	0.67	-0.20**	1.66***
months	(-11.18)	(0.000)	(-7.86)	(7.57)	(1.57)	(-2.43)	(6.76)

The default spread coefficients indicate heterogeneity between the equal-weighted and value-weighted ESG momentum strategies. For the equal-weighted portfolios, the coefficients are negative and statistically significant. However, for the value-weighted portfolios, the coefficients are positive and, except for the 36-month specification, statistically significant. A possible interpretation is that widening default spreads affect small stocks, which receive a larger weight in the equal-weighted portfolio, differently than large stocks. There exists literature showing that small firms are more strongly affected by changes in credit market conditions than large firms. For example, Perez-Quiros & Timmermann (2000) find that small firms display a higher sensitivity of their expected stock returns with respect to variables that measure credit market conditions such as the default spread. However, the negative coefficients on default spreads shouldn't be interpreted as implying that small stocks earn lower returns when credit spreads widen. Rather, they indicate that an increase in default spreads tends to reduce the return spread between the high- and low ESG momentum portfolios in the equal-weighted strategy. Further research could investigate the interaction between stock returns, size, default spreads and ESG score improvements in more detail to provide a deeper understanding of this relationship.

4.2.5 Business Cycle Analysis Summary

Overall, the results of the business cycle analysis suggest that ESG momentum payoffs in European equities are related to macroeconomic conditions, although the evidence is more supportive than definitive. Hence, research question **Q2** can be answered with a qualified yes: common macroeconomic variables seem to be associated with a meaningful part of ESG momentum payoffs, although the evidence is sensitive to specification choices such as the rolling estimation window length and weighting scheme. Regarding **Q3**, the results indicate that ESG momentum payoffs are related to the business cycle through their exposure to macroeconomic conditions. Both equal-weighted and value-weighted strategies show positive cumulative payoffs from June 2016 to September 2025, with the strongest payoffs occurring during the COVID-19 recession window. However, payoffs are also positive, although mostly insignificant, in both non-recession sub periods. Stocks with high ESG momentum generally exhibit higher macro-predicted returns than stocks with low ESG momentum, and once this predictable component is removed, the remaining payoff is small and often negative, especially in value-weighted portfolios. Direct regressions of ESG momentum payoffs on lagged macroeconomic variables likewise suggest that dividend yield, short-term yield, term spread, and default spread help explain a meaningful share of the strategy's payoffs. This pattern is consistent with, though it does not by itself establish, the view that ESG improvements are valued more strongly when discount rates are high and risk reduction becomes more valuable.

5 Conclusion

This thesis examined whether ESG momentum is related to stock returns in European equity markets and whether ESG momentum payoffs are linked to macroeconomic variables and business cycle conditions. The analysis combined pooled OLS, random effects, fixed effects, and two-step difference GMM with a business cycle analysis based on CEPR-EABCN recession dates, dividend yield, short-term yield, term spread, and default spread. Research question **Q1** asks whether ESG momentum is positively associated with quarterly stock returns among STOXX Europe 600 stocks. The panel results suggest that ESG score levels are negatively related to returns, while ESG momentum appears to be positively associated with returns. However, significance is sensitive to model specification, instrument lags, and sample period. Overall, the sign pattern documented by Escobar-Saldivar et al. (2025) is also visible among European equities, so Q1 can be answered with a qualified yes. **Q2** asks whether the macro variables explain ESG momentum payoffs. This question can also be answered with a qualified yes. The variables explain a meaningful part of the positive payoff component, and after removing this component, the average unexplained component becomes often negative. However, significance is consistent only for value-weighted payoffs. **Q3** asks how ESG momentum payoffs relate to the business cycle. The evidence suggests that this relation exists through exposure to macroeconomic conditions. Payoffs peak during the COVID-19 pandemic; high ESG momentum portfolios show higher macro-predicted returns than low ESG momentum portfolios, while residual payoffs shrink or turn negative after macro adjustment. Direct payoff regressions further show that payoffs are higher when dividend yields and short-term rates are high and when the yield curve is flatter or inverted, while the relation to default spreads depends on the weighting scheme. If ESG improvements signal lower risk, these signals may matter more under such conditions. Since the sample contains only one short recession period and results vary by weighting scheme and training window, the findings are suggestive rather than stable. Overall, the results are consistent with the interpretation that ESG levels and ESG momentum capture different economic mechanisms: ESG levels may reflect lower risk and lower required returns, whereas ESG momentum may capture repricing while new ESG-related information is incorporated into prices, with the evidence suggesting that this repricing is partly related to macroeconomic conditions such as discount rates, monetary conditions, and a default-risk proxy. The thesis covers only June 2016 to September 2025 and includes only one short recession, limiting generalizability. The GMM results are sensitive to instrument lags and sub-periods. Finally, the thesis relies on Refinitiv ESG scores, while prior work raises concerns about historical rewriting. Future research could use longer samples, additional recessions, alternative ESG providers, and examine interactions between firm size, macroeconomic variables, and ESG momentum.

A Appendix

A.1 Complete Results Of Main Specifications POLS, FE, RE, GMM

Table 9: Complete Results of Main Specifications POLS, FE, RE, GMM

This table presents the complete set of estimated coefficients for the period from Jun 2016 to Sep 2025 of the multi-factor models in Equations 1 and 8 obtained by performing pooled OLS (POLS), fixed effects (FE), random effects (RE), and two-step difference generalized method of moments (GMM) as described in chapters 2.3.1.1 and 2.3.1.2. Independent variables are winsorized at the 0.5% and 99.5% level. Entity clustered t-statistics are shown in parenthesis for POLS, FE, and RE. Windmeijer-corrected z-statistics are shown in parenthesis for GMM. *, **, *** show statistical significance at the 10%, 5%, and 1% level.

	POLS	FE	RE	GMM (1)
Observations	20,184	20,184	20,184	17,203
R-Squared (Within)	0.6510	0.3246	0.6510	
R-Squared (Between)	0.4323	0.2007	0.4323	
R-Squared (Overall)	0.4418	0.1657	0.4418	
R_{q-1}				0.0167 (0.48)
Beta	0.0031*** (2.85)	0.0034*** (2.62)	0.0031*** (2.85)	0.0054 (0.94)
Size	0.0075*** (11.97)	0.0358*** (8.19)	0.0075*** (11.97)	0.0306 (0.76)
Value	-0.0155*** (-6.64)	-0.0421*** (-5.69)	-0.0155*** (-6.64)	-0.0813** (-2.08)
Prof	-0.0094*** (-3.41)	-0.0106** (-2.15)	-0.0094*** (-3.41)	0.057 (1.52)
Inv	0.0101 (1.12)	0.0082 (0.93)	0.0101 (1.12)	-0.5012 (-1.08)
Price Mom	0.3343*** (61.49)	0.3046*** (46.50)	0.3343*** (61.49)	-0.0453 (-1.07)
ESG	-0.0001 (-1.49)	-0.0005*** (-3.78)	-0.0001 (-1.49)	-0.0019** (-2.33)
ESG Mom	0.0001 (0.53)	0.0002* (1.68)	0.0001 (0.53)	0.0012* (1.85)
Constant	-0.17*** (-11.12)		-0.17*** (-11.12)	
Industry.BasicMaterials	<i>Omitted</i>		<i>Omitted</i>	
Industry.ConsumerDisc	-0.005** (-2.22)		-0.005** (-2.22)	
Industry.ConsumerStaples	-0.01*** (-3.96)		-0.01*** (-3.96)	
Industry.Energy	-0.002 (-0.62)		-0.002 (-0.62)	

Table 9 Continued

	POLS	FE	RE	GMM (1)
Industry.Financials	0.004 (1.52)		0.004 (1.52)	
Industry.HealthCare	-0.009*** (-3.51)		-0.009*** (-3.51)	
Industry.Industrials	0.001 (0.36)		0.001 (0.36)	
Industry.RealEstate	0.004 (1.48)		0.004 (1.48)	
Industry.Technology	-0.007** (-2.11)		-0.007** (-2.11)	
Industry.Telecommunications	-0.006** (-2.21)		-0.006** (-2.21)	
Industry.Utilities	-0.004 (-1.48)		-0.004 (-1.48)	
Country.Austria	<i>Omitted</i>		<i>Omitted</i>	
Country.Belgium	-0.007 (-1.32)		-0.007 (-1.32)	
Country.Bermuda	-0.008** (-2.15)		-0.008** (-2.15)	
Country.Cyprus	0.031*** (8.02)		0.031*** (8.02)	
Country.CzechRepublic	0.0003 (0.06)		0.0003 (0.06)	
Country.Denmark	-0.007 (-1.62)		-0.007 (-1.62)	
Country.FaroeIslands	0.003 (0.68)		0.003 (0.68)	
Country.Finland	-0.006 (-1.48)		-0.006 (-1.48)	
Country.France	-0.008** (-2.13)		-0.008** (-2.13)	
Country.Germany	-0.008** (-1.99)		-0.008** (-1.99)	
Country.Hungary	-0.092*** (-18.64)		-0.092*** (-18.64)	
Country.Ireland	-0.008* (-1.83)		-0.008* (-1.83)	
Country.IsleOfMan	-0.027*** (-6.76)		-0.027*** (-6.76)	
Country.Israel	-0.016*** (-3.41)		-0.016*** (-3.41)	
Country.Italy	-0.004 (-0.89)		-0.004 (-0.89)	
Country.Jersey	-0.007 (-1.35)		-0.007 (-1.35)	
Country.Kazakhstan	-0.008* (-1.77)		-0.008* (-1.77)	
Country.Luxembourg	-0.001 (-0.26)		-0.001 (-0.26)	

Table 9 Continued

	POLS	FE	RE	GMM (1)
Country.Malta	0.001 (0.3)		0.001 (0.3)	
Country.Mexico	0.02*** (4.45)		0.02*** (4.45)	
Country.Netherlands	-0.003 (-0.69)		-0.003 (-0.69)	
Country.Norway	-0.005 (-1.12)		-0.005 (-1.12)	
Country.Poland	0.004 (0.5)		0.004 (0.5)	
Country.Portugal	-0.004 (-1.1)		-0.004 (-1.1)	
Country.SouthAfrica	-0.06 (-1.2)		-0.06 (-1.2)	
Country.Spain	-0.007 (-1.58)		-0.007 (-1.58)	
Country.Sweden	-0.008** (-2.01)		-0.008** (-2.01)	
Country.Switzerland	-0.011*** (-2.75)		-0.011*** (-2.75)	
Country.UnitedKingdom	-0.007* (-1.88)		-0.007* (-1.88)	
Country.UnitedStates	-0.004 (-1.14)		-0.004 (-1.14)	
Time.Jun2016	<i>Omitted</i>		<i>Omitted</i>	
Time.Sep2016	0.065*** (7.87)		0.065*** (7.87)	
Time.Dec2016	0.061*** (7.76)		0.061*** (7.76)	
Time.Mar2017	0.056*** (8.33)		0.056*** (8.33)	
Time.Jun2017	-0.032*** (-5.01)		-0.032*** (-5.01)	-0.0851*** (-3.85)
Time.Sep2017	0.036*** (5.43)		0.036*** (5.43)	-0.0753** (-2.35)
Time.Dec2017	0.029*** (4.4)		0.029*** (4.4)	-0.0824*** (-2.87)
Time.Mar2018	-0.029*** (-4.5)		-0.029*** (-4.5)	-0.1111*** (-8.42)
Time.Jun2018	0.049*** (6.5)		0.049*** (6.5)	-0.0590** (-2.19)
Time.Sep2018	0.013* (1.86)		0.013* (1.86)	-0.0874*** (-3.19)
Time.Dec2018	-0.085*** (-12.79)		-0.085*** (-12.79)	-0.2158*** (-8.02)
Time.Mar2019	0.166*** (23.3)		0.166*** (23.3)	0.0687*** (5.83)
Time.Jun2019	0.018*** (2.78)		0.018*** (2.78)	-0.0674*** (-4.41)

Table 9 Continued

	POLS	FE	RE	GMM (1)
Time.Sep2019	0.036*** (5.08)		0.036*** (5.08)	-0.0721** (-2.41)
Time.Dec2019	0.055*** (7.41)		0.055*** (7.41)	-0.0009 (-0.05)
Time.Mar2020	-0.236*** (-29.82)		-0.236*** (-29.82)	-0.2878*** (-16.01)
Time.Jun2020	0.221*** (24.7)		0.221*** (24.7)	0.0942*** (4.69)
Time.Sep2020	0.051*** (6.57)		0.051*** (6.57)	-0.0423 (-1.60)
Time.Dec2020	0.107*** (10.98)		0.107*** (10.98)	0.0636** (2.46)
Time.Mar2021	0.039*** (5.39)		0.039*** (5.39)	0.0079 (0.53)
Time.Jun2021	0.013** (1.99)		0.013** (1.99)	-0.0219 (-1.16)
Time.Sep2021	-0.038*** (-5.45)		-0.038*** (-5.45)	-0.0776*** (-3.30)
Time.Dec2021	0.064*** (9.48)		0.064*** (9.48)	-0.0179 (-0.94)
Time.Mar2022	-0.041*** (-5.38)		-0.041*** (-5.38)	-0.1491*** (-10.67)
Time.Jun2022	-0.097*** (-13.9)		-0.097*** (-13.9)	-0.1995*** (-12.16)
Time.Sep2022	-0.02*** (-2.81)		-0.02*** (-2.81)	-0.1450*** (-6.51)
Time.Dec2022	0.142*** (17.76)		0.142*** (17.76)	0.0406** (2.11)
Time.Mar2023	0.068*** (8.7)		0.068*** (8.7)	0.0187 (1.04)
Time.Jun2023	-0.003 (-0.39)		-0.003 (-0.39)	-0.0879*** (-3.33)
Time.Sep2023	-0.003 (-0.37)		-0.003 (-0.37)	-0.1068*** (-3.83)
Time.Dec2023	0.104*** (13.39)		0.104*** (13.39)	0.0117 (0.45)
Time.Mar2024	0.035*** (4.64)		0.035*** (4.64)	-0.0463* (-1.78)
Time.Jun2024	-0.035*** (-5.28)		-0.035*** (-5.28)	-0.0856*** (-4.10)
Time.Sep2024	0.049*** (6.58)		0.049*** (6.58)	-0.0305 (-1.11)
Time.Dec2024	-0.02*** (-2.9)		-0.02*** (-2.9)	-0.1263*** (-4.66)
Time.Mar2025	0.033*** (4.1)		0.033*** (4.1)	-0.0388** (-2.41)
Time.Jun2025	0.04*** (5.23)		0.04*** (5.23)	-0.0394 (-1.47)
Time.Sep2025	0.02*** (2.99)		0.02*** (2.99)	-0.0874*** (-2.77)

Table 9 Continued

Notes on GMM:
<i>Hansen p-value: 0.231; AR(1) p-value: 0.000; AR(2) p-value: 0.183</i>
<i>Instruments are collapsed. Instrument Count: 52</i>
<i>Lags used: $r_{t-1} = L.2-L.3$; Price Momentum = L.4-L.5; All others = L.3</i>

A.2 Complete Results Of Robustness Specifications of GMM

Table 10: Complete Results of Robustness Specifications of GMM

This table presents the complete set of estimated coefficients for six different specifications of the two-step difference generalized method of moments model in Equation 8. Any specification changes made to the base model GMM (1) presented in table 3, are highlighted in bold. Independent variables are winsorized at the 0.5% and 99.5% level. Windmeijer-corrected z-statistics are shown in parenthesis. *, **, *** show statistical significance at the 10%, 5%, and 1% level.

	GMM (2)	GMM (3)	GMM (4)	GMM (5)	GMM (6)	GMM (7)
Sample Period	06/2016 - 09/2025	06/2016 - 09/2025	06/2016 - 09/2025	06/2016 - 06/2019	09/2019 - 09/2022	12/2022 - 09/2025
Lags:						
R_{q-1}	2-3	2-3	2-3	2-3	2-3	2-3
Beta	4-4	3-3	3-3	3-3	3-3	3-3
Size	4-4	3-3	3-3	3-3	3-3	3-3
Value	4-4	3-3	3-3	3-3	3-3	3-3
Prof	4-4	3-3	N/A	3-3	3-3	3-3
Inv	4-4	3-3	3-3	3-3	3-3	3-3
Price Mom	6-6	3-3	3-3	4-5	4-5	4-5
ESG	4-4	3-3	3-3	3-3	3-3	3-3
ESG Mom	4-4	3-3	3-3	3-3	3-3	3-3
R_{q-1}	0.0114 (0.35)	0.0165 (0.49)	0.0216 (0.63)	0.0147 (0.13)	-0.0843 (-0.80)	0.959 (0.60)
Beta	0.027*** (3.18)	0.0085 (1.46)	0.0066 (1.11)	0.0052 (0.24)	0.0342 (0.83)	0.0050 (0.38)
Size	0.008 (0.19)	0.0402 (1.04)	0.0510 (1.26)	0.0754 (0.39)	0.0252 (0.06)	-0.0119 (-0.07)
Value	-0.0809** (-2.19)	-0.0499 (-1.32)	-0.0569 (-1.43)	-0.1913 (-0.67)	-0.6564* (-1.75)	-0.2211 (-1.56)
Prof	0.0226 (0.57)	0.0400 (1.30)	-0.0119 (-0.96)	-0.0521 (-0.23)	0.3996** (2.01)	0.1823 (1.39)
Inv	0.0285 (0.11)	-0.5163 (-1.10)	-0.5936 (-1.22)	0.5766 (1.00)	-2.3360** (-2.23)	0.7481 (0.36)
Price Mom	-0.0317 (0.85)	-0.0373 (-0.88)	-0.0486 (-1.12)	-0.0396 (-0.29)	0.0261 (0.17)	-0.2145 (-1.03)
ESG	-0.0010 (-1.56)	-0.0019** (-2.38)	-0.0020** (-2.53)	0.0001 (0.02)	-0.0153** (-2.01)	-0.0106 (-1.51)
ESG Mom	0.0003 (0.28)	0.0012* (1.75)	0.0013* (1.89)	0.0020 (1.58)	0.0021 (0.59)	0.0002 (0.007)
Time.Mar2017		0.0579** (1.97)	0.0599** (2.01)			
Time.Jun2017		-0.0299** (-2.48)	-0.0307** (-2.54)	-0.0325 (-1.26)		
Time.Sep2017		-0.0176** (-1.98)	-0.0205** (-2.23)	-0.0044 (-0.11)		

Table 10 Continued

	GMM (2)	GMM (3)	GMM (4)	GMM (5)	GMM (6)	GMM (7)
Time.Dec2017	-0.0139** (-1.98)	-0.025*** (-2.93)	-0.028*** (-3.22)	-0.0270 (-0.92)		
Time.Mar2018	-0.062*** (-4.55)	-0.055*** (-2.79)	-0.054*** (-2.73)	-0.088*** (-5.06)		
Time.Jun2018	0.0119 (1.44)	-0.0021 (-0.21)	-0.0036 (-0.36)	-0.0021 (-0.06)		
Time.Sep2018	-0.0151* (-1.93)	-0.031*** (-3.51)	-0.033*** (-3.70)	-0.0315 (-0.82)		
Time.Dec2018	-0.149*** (-12.50)	-0.160*** (-15.41)	-0.161*** (-15.59)	-0.149*** (-3.75)		
Time.Mar2019	0.108*** (5.88)	0.126*** (4.14)	0.130*** (4.18)	0.067** (2.09)		
Time.Jun2019	-0.0157 (-1.21)	-0.0121 (-0.74)	-0.0109 (-0.66)	-0.0374 (-1.22)		
Time.Sep2019	-0.0011 (-0.13)	-0.0178* (-1.75)	-0.0191* (-1.87)			
Time.Dec2019	0.057*** (5.42)	0.052*** (3.44)	0.053*** (3.44)			
Time.Mar2020	-0.253*** (-10.96)	-0.239*** (-7.68)	-0.233*** (-7.27)			
Time.Jun2020	0.147*** (9.92)	0.146*** (7.93)	0.146*** (7.91)			
Time.Sep2020	0.019 (1.63)	0.009 (0.83)	0.007 (0.67)		-0.147*** (-3.85)	
Time.Dec2020	0.124*** (10.24)	0.114*** (7.88)	0.112*** (7.77)		-0.0931** (-2.31)	
Time.Mar2021	0.055*** (3.83)	0.060*** (2.79)	0.057*** (2.73)		-0.076 (-1.51)	
Time.Jun2021	0.036*** (2.80)	0.030 (1.58)	0.026 (1.36)		-0.136** (-2.12)	
Time.Sep2021	-0.0133 (-1.09)	-0.0266 (-1.56)	-0.0306* (-1.82)		-0.216*** (-3.15)	
Time.Dec2021	0.039*** (2.71)	0.034* (1.69)	0.030 (1.49)		-0.112 (-1.28)	
Time.Mar2022	-0.108*** (-5.71)	-0.097*** (-3.60)	-0.097*** (-3.57)		-0.168** (-2.42)	
Time.Jun2022	-0.150*** (-9.95)	-0.150*** (-7.58)	-0.149*** (-7.39)		-0.209*** (-5.74)	
Time.Sep2022	-0.089*** (-6.11)	-0.097*** (-6.16)	-0.095*** (-5.88)		-0.144*** (-4.22)	
Time.Dec2022	0.094*** (6.73)	0.089*** (5.09)	0.090*** (5.07)			
Time.Mar2023	0.068*** (4.87)	0.066*** (3.31)	0.066*** (3.28)			
Time.Jun2023	-0.027** (-2.29)	-0.041*** (-3.13)	-0.043*** (-3.27)			

Table 10 Continued

	GMM (2)	GMM (3)	GMM (4)	GMM (5)	GMM (6)	GMM (7)
Time.Sep2023	-0.041*** (-3.60)	-0.056*** (-4.84)	-0.061*** (-5.00)			
Time.Dec2023	0.072*** (6.06)	0.060*** (4.07)	0.058*** (3.90)			0.111*** (6.77)
Time.Mar2024	0.015 (1.24)	0.002 (0.12)	-0.001 (-0.08)			0.052** (2.46)
Time.Jun2024	-0.033** (-2.44)	-0.039** (-2.16)	-0.040** (-2.19)			0.018 (0.60)
Time.Sep2024	0.033** (2.48)	0.017 (1.08)	0.015 (0.90)			0.085*** (2.96)
Time.Dec2024	-0.061*** (-5.08)	-0.078*** (-5.98)	-0.081*** (-6.17)			-0.034 (-1.46)
Time.Mar2025	0.014 (0.87)	0.010 (0.39)	0.010 (0.40)			0.029 (0.45)
Time.Jun2025	0.032** (2.54)	0.010 (0.67)	0.007 (0.48)			0.063** (2.53)
Time.Sep2025	-0.013 (-1.12)	-0.037*** (-2.76)	-0.041*** (-2.97)			0.005 (0.27)
Instrument Count	42	45	45	20	20	19
P-values:						
Hansen	0.734	0.133	0.172	0.148	0.139	0.907
AR(1)	0.000	0.000	0.000	0.003	0.009	0.004
AR(2)	0.238	0.085*	0.118	0.996	0.163	0.032**

A.3 Detailed Data Sources of Variables

Table 11: Data Sources of Variables

This table presents the data sources for all variables used in this study. For any calculated variable, the data sources of the individual components are provided. For variables obtained from Refinitiv, data items are provided in parenthesis. The variable definitions and calculation methodology are provided in chapter 2.3.

Variable	Data Source
Stock Returns	Stock closing price: Refinitiv (TR.PriceClose)
Beta	Index closing price: Refinitiv (TR.PriceClose)
Size	Refinitiv (TR.CompanyMarketCap(ShType=OUT))
Value	Refinitiv (TR.PriceToBVPerShare)
Profitability	Refinitiv (TR.OperatingIncome) and Refinitiv (TR.TotalEquity)
Investment	Refinitiv (TR.TotalAssetsReported)
Price Momentum	Stock closing price: Refinitiv (TR.PriceClose)
ESG Score & ESG Momentum	Refinitiv (TR.TRESGScore)
Dividend Yield	Index & Stock Prices: Refinitiv (TR.PriceClose); Free-Float Market Capitalization: Refinitiv (TR.CompanyMarketCap(ShType=FFL)); Dividend Payments: Refinitiv (TR.DivAdjustedGross); Ex dividend date: Refinitiv (TR.DivExDate); Dividend Type: Refinitiv (TR.DivPaymentType)
Yield	https://data.ecb.europa.eu/data/datasets/YC/YC.B.U2.EUR.4F.G_N_A.SV_C_YM.SR_3M
Term Spread	10-year spot rate: https://data.ecb.europa.eu/data/datasets/YC/YC.B.U2.EUR.4F.G_N_A.SV_C_YM.SR_10Y 3-month spot rate: https://data.ecb.europa.eu/data/datasets/YC/YC.B.U2.EUR.4F.G_N_A.SV_C_YM.SR_3M
Default Spread	BAA Bonds: https://fred.stlouisfed.org/series/BAA AAA Bonds: https://fred.stlouisfed.org/series/AAA
Svensson parameters:	
Beta 0	https://data.ecb.europa.eu/data/datasets/YC/YC.B.U2.EUR.4F.G_N_A.SV_C_YM.BETA0
Beta 1	https://data.ecb.europa.eu/data/datasets/YC/YC.B.U2.EUR.4F.G_N_A.SV_C_YM.BETA1
Beta 2	https://data.ecb.europa.eu/data/datasets/YC/YC.B.U2.EUR.4F.G_N_A.SV_C_YM.BETA2
Beta 3	https://data.ecb.europa.eu/data/datasets/YC/YC.B.U2.EUR.4F.G_N_A.SV_C_YM.BETA3
Tau 1	https://data.ecb.europa.eu/data/datasets/YC/YC.B.U2.EUR.4F.G_N_A.SV_C_YM.TAU1
Tau 2	https://data.ecb.europa.eu/data/datasets/YC/YC.B.U2.EUR.4F.G_N_A.SV_C_YM.TAU2
Fama-French Factor Returns	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
EABCN Business Cycle Dates	https://eabcn.org/dbc/graphs

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